



MASTER'S THESIS

**Foundation-Model-Guided Zero-Shot
Synthesis of Human-Scene Interaction**

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Declaration

I declare that this thesis has been completed independently and that no sources or aids other than those listed have been used.

Munich, 20.07.2026

Place, Date

A handwritten signature in black ink, featuring a large, stylized 'U' or 'W' shape with a horizontal line extending to the left and a vertical line extending downwards to the right.

Signature

Abstract

Generating a human who interacts convincingly with a real 3D scene is difficult because the input is usually semantic, while the output must satisfy geometry. A hand should meet the correct object surface, feet should support the body, and the body should not pass through nearby furniture or the room itself. This thesis studies a zero-shot pipeline for static human-scene interaction synthesis from a single RGB view, a ScanNet++ scene mesh, and a language instruction.

The method uses foundation models where their strengths are most useful, but keeps the final result tied to metric 3D geometry. First, a vision-language model converts the instruction into a Scene Interaction Graph that names the target object, relevant human body parts, and intended contact edges. An image editing model then inserts a plausible human into the scene, producing a 2D reference for the interaction. A second contact-inpainting step estimates object-surface contact masks for the body parts specified by the graph. These masks are projected from the image view onto the visible scene mesh and converted into 3D contact targets. Starting from an SMPL-X body initialized with GVHMR, the final stage optimizes pose, orientation, translation, and scale using contact, depth ordering, scene collision, self-intersection, height, and pose-prior terms.

The pipeline is evaluated on real indoor scenes using physical contact distances, non-collision metrics, render-based semantic consistency, and vision-language model verification. The result is a practical framework for turning text-level interaction intent into a geometrically grounded 3D human placement without task-specific training.

Keywords: human-scene interaction, foundation models, zero-shot synthesis, SMPL-X, contact optimization

Acknowledgments

Thank advisors, supervisors, collaborators, funding sources, friends, and family here. Keep this section personal and direct.

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List of Abbreviations

HOI	Human-Object Interaction
ML	Machine Learning
RGB-D	Red, Green, Blue, and Depth

Chapter 1

Introduction

This chapter motivates zero-shot static 3D human-scene interaction synthesis from language, a calibrated RGB image, and a reconstructed scene mesh. It places the thesis in relation to learned HSI models, scene-constrained body optimization, and recent foundation-model-based systems, then states the problem formulation, approach, and contributions.

1.1 Why Human-Scene Interaction Matters

Indoor scenes are built around human action. Chairs are placed so that people can sit, tables so that people can work or eat, cabinets so that objects can be stored and retrieved, and handles so that drawers, doors, and appliances can be opened. The meaning of these objects is therefore not exhausted by their geometry or category labels. A chair is not only a shape occupying space; it affords support for a seated body. A drawer is not only a box-like structure inside a cabinet; it has a front face and a handle that invite a particular hand motion. A room becomes meaningful because human bodies can stand, reach, grasp, lean, sit, lie, and move through it.

This distinction is central to 3D scene understanding. Modern reconstruction methods can produce increasingly accurate models of indoor environments, including camera poses, dense geometry, object surfaces, and scene layout. However, a reconstructed scene by itself remains a passive description of the environment. It tells us where surfaces are, but not which surfaces can support the body. It shows object boundaries, but not which regions are functional for a task. It captures the state of a room, but not the ways in which a person could physically use that room. For applications that require human-centered reasoning, this missing layer is fundamental.

Human-scene interaction provides this missing layer by connecting scene structure to bodily action. Instead of asking only what objects are present, it asks how a person can engage with them. Where should the feet be placed when reaching for a cabinet? Which part of

a sofa should support the pelvis and back during sitting? Where should the hand make contact when opening a drawer? What body orientation is plausible when lifting a trash bin? These questions require reasoning jointly about object function, human body parts, contact, balance, reachability, and spatial constraints. They cannot be answered by object recognition or scene reconstruction alone.

This capability is important for a broad class of downstream systems. In embodied AI and robotics, an agent must reason not only about object locations, but also about how humans act around those objects and which parts of the scene are functionally relevant. In augmented and virtual reality, inserted humans must appear grounded in the environment rather than floating near objects or intersecting with them. In animation, synthetic data generation, and digital twins, realistic human activity requires bodies that are placed according to the affordances of the scene. In human-centered scene analysis, the presence or absence of plausible interactions can reveal whether a reconstructed environment is usable, accessible, or semantically coherent.

The challenge is therefore not simply to add a human body to a 3D scene. A visually plausible body can still be physically or functionally wrong: a seated person may hover above a chair, a hand may reach the side of a cabinet instead of the drawer handle, or a body may intersect with furniture while appearing reasonable from one camera view. Human-scene interaction requires a stronger form of consistency. The body must be semantically aligned with the requested action, geometrically aligned with the scene, and physically plausible with respect to contact and collision.

This thesis studies human-scene interaction synthesis as a step toward scene representations that are not only geometric, but actionable. The goal is to synthesize human bodies that express how a real indoor scene can be used. By grounding human pose in scene affordances, object function, and physical contact, HSI synthesis moves beyond static reconstruction toward a more complete understanding of indoor environments as spaces for human activity.

1.2 From Reconstructed Geometry to Interaction Synthesis

A 3D reconstruction converts visual observations of a room into a metric representation of space. It provides camera poses, scene geometry, object surfaces, and the spatial layout of the environment. This representation is essential for grounding perception in real-world coordinates: it tells us where the floor is, how far a chair is from a table, where a cabinet is located, and which surfaces are visible or occluded. However, reconstruction alone does not specify how a human body should be arranged in the scene. Geometry describes the shape of the environment, but interaction requires reasoning about the relationship between that geometry and the body.

This gap becomes apparent even for simple activities. To synthesize a person sitting on a chair, it is not enough to know that a chair exists. The body must be placed so that the pelvis and thighs are supported by the seat, the feet remain near the floor, and the torso has a plausible orientation relative to the chair. To synthesize a person opening a drawer, it is not enough to locate the cabinet. The hand should reach the drawer pull or handle, while the body should stand at a reachable distance and face the object in a plausible way. To synthesize a person lifting a trash bin, the hands, arms, torso, and feet must form a coherent configuration around the object. In each case, the interaction is defined not only by the object category, but by the spatial and functional relation between body parts and scene parts.

Human-scene interaction synthesis addresses this problem by generating a human body configuration that is jointly consistent with an action and a 3D scene. Unlike image-based human insertion, the goal is not only to make a rendered view appear plausible. The synthesized body must exist in the same coordinate frame as the scene and must satisfy constraints that are meaningful in 3D: contact should occur at appropriate locations, the body should have a plausible scale and orientation, and severe interpenetration with the environment should be avoided. The output is therefore a scene-grounded human state, not simply a visual overlay.

This distinction separates interaction synthesis from several related tasks. Object detection identifies what is present in the scene, but it does not determine how the object can be used. Human pose estimation recovers a body configuration from observations, but it usually assumes that the person is already present. Text-to-image generation can produce visually convincing depictions of people interacting with objects, but the result is not necessarily metrically consistent with a reconstructed scene. Human-scene interaction synthesis combines aspects of all three problems while adding a stronger requirement: the generated body must be semantically aligned with the requested action and physically grounded in the 3D environment.

The difficulty is that interaction is structured. A single instruction often implies a set of relations among the action, the object, the relevant object part, the participating body parts, and the supporting scene surfaces. Sitting involves support relations between the lower body and a seat. Opening involves a reaching or grasping relation between a hand and a functional element. Lifting involves coordinated contacts between the hands and the object, together with a stable stance on the floor. These relations are not directly given by the reconstruction. They must be inferred from the task and then realized as a plausible body placement in the scene.

Interaction synthesis can therefore be viewed as the step from a passive scene model to an actionable scene model. A passive model answers geometric questions such as where surfaces are and how objects are arranged. An actionable model supports questions about what a human can do in the environment, where the body should be placed, which body parts

should contact which scene regions, and whether the resulting configuration is physically credible. This thesis focuses on this transition: synthesizing a human body that does not merely appear inside a reconstructed room, but uses the room in a way that is consistent with language, geometry, and physical contact.

1.3 The Zero-Shot Challenge

Human-scene interaction synthesis becomes especially difficult when the system is expected to operate in a zero-shot setting. In a closed setting, the model may assume a fixed set of object categories, a fixed set of actions, or training examples that closely resemble the test scenes. Under these assumptions, the problem can be approached as learning regularities from paired examples of humans interacting with objects. The model can learn, for instance, that people sit on chairs, stand on floors, touch tables with their hands, or lie on beds. However, such regularities are only reliable when the test interaction lies within the distribution of the training data.

Real indoor scenes do not satisfy this assumption. They contain diverse object instances, cluttered layouts, partial visibility, unusual viewpoints, and functional elements that vary widely in appearance. A drawer handle may be horizontal, vertical, recessed, metallic, wooden, large, small, or partially occluded. A chair may be an office chair, dining chair, stool, sofa, or bench. Even when the object category is familiar, the region relevant to an interaction may differ across instances. The same action phrase can therefore require different geometric solutions depending on the scene. ‘Open the drawer’ requires the hand to reach the handle of a particular drawer in a particular cabinet. ‘Sit on the chair’ requires the body to find a support surface of the correct height and orientation. ‘Lift the bin’ requires a graspable region on the visible object and a stable full-body configuration around it.

This variability exposes the limitation of relying only on supervised 3D human-scene interaction data. Paired HSI data is expensive to collect because it requires reconstructed scenes, accurate human body fits, object annotations, action labels, contact information, and enough diversity in body shapes, object instances, and room layouts. As a result, existing datasets cover only a small subset of the possible interactions that occur in indoor environments. Learned HSI models can produce strong results for common interactions represented in their training sets, but they often depend on closed vocabularies, predefined object categories, known target objects, or interaction patterns observed during training. These constraints make it difficult to synthesize interactions for arbitrary language instructions in previously unseen reconstructed scenes.

Recent work has therefore explored the use of large vision-language, image-generation, and video-generation models as sources of interaction knowledge. These models have seen

broad visual and linguistic evidence of people using objects, and they can often suggest plausible body poses, object affordances, and interaction layouts without being trained directly on 3D HSI datasets. This has motivated zero-shot and training-free approaches to human-scene interaction synthesis, where foundation models provide semantic or visual priors and 3D geometry provides metric grounding. Methods such as GenZI, FunHSI, and ZeroHSI show that this direction is promising because interaction knowledge can be transferred from large-scale visual models to 3D scenes without requiring paired 3D interaction supervision.

However, zero-shot HSI is not solved by generating a plausible image of a person interacting with an object. A generated image may express the intended action visually, but it does not guarantee that the body is correctly placed in the reconstructed scene. The hands may appear near an object in image space while missing the corresponding 3D surface. The body may look plausible from one camera view while penetrating furniture or floating above the floor. The generated interaction may also contact the wrong part of the object: the side of a cabinet instead of the drawer handle, the rim of a bin instead of a graspable side, or the back of a chair instead of the seat. The challenge is therefore to convert broad interaction priors into scene-specific, metrically valid human-body configurations.

The zero-shot setting thus requires solving several coupled questions. Given an open-vocabulary instruction, the system must infer which scene object is relevant, which part of that object affords the action, which human body parts should participate, where supporting contacts should occur, and how the full body should be arranged in 3D. These questions cannot be answered independently. The choice of contact region affects the pose; the pose affects the feasible body location; the body location affects reachability, balance, and collision; and the scene geometry constrains all of these decisions. Zero-shot HSI is therefore not simply a problem of text-conditioned pose generation, but one of grounding an inferred interaction structure in a specific reconstructed environment.

This thesis addresses the zero-shot challenge in the static 3D setting. Rather than assuming a fixed action vocabulary or paired 3D interaction examples for every task, it treats the instruction as an open-ended description of how a person should use the scene. The goal is to infer the relevant interaction structure and synthesize a single SMPL-X body that realizes that structure in the original scene coordinate frame. This formulation keeps the problem narrower than full dynamic interaction generation, but it preserves the central difficulty of zero-shot HSI: the system must transform language and scene observations into a physically plausible human-scene contact configuration without relying on task-specific 3D supervision.

1.4 Body-Part–Scene-Part Grounding

The previous sections framed human-scene interaction synthesis as more than placing a plausible body near a relevant object. The remaining question is what structure should connect the language instruction, the human body, and the scene. A natural first answer is to reason at the object level: a person sits on a chair, opens a drawer, touches a table, or lifts a bin. Object-level reasoning is useful, but it is not precise enough for interaction synthesis. Most interactions are not defined by whole objects. They are defined by relations between specific parts of the human body and specific regions of the scene.

Consider the instruction ‘a person opening a drawer.’ The target object may be a cabinet or drawer unit, but the interaction itself is concentrated at the drawer handle or front edge. The relevant human part is usually a hand, not the entire body. At the same time, the rest of the body is not arbitrary: the feet must remain supported by the floor, the torso must be oriented toward the cabinet, and the arm must be arranged so that the hand can reach the functional part. Similarly, ‘a person sitting on a chair’ primarily requires support between the lower body and the seat, but also implies a plausible relation between the feet and the floor and between the torso and the chair orientation. These examples show that an interaction is best understood as a set of body-part–scene-part relations.

This thesis adopts that view. Instead of treating the human body as a single object and the scene as a collection of object categories, it represents an interaction through participating body regions, scene regions, and contact relations between them. The human body is decomposed into interaction-relevant parts, such as hands, arms, feet, pelvis, thighs, back, and torso. These parts correspond to regions on the SMPL-X surface that commonly participate in physical interactions. On the scene side, the relevant regions may be broad support surfaces, such as floors, seats, beds, and tabletops, or smaller functional elements, such as handles, knobs, switches, faucets, and drawer pulls. The task of HSI synthesis is then to arrange the body so that the appropriate body regions are grounded to the appropriate scene regions.

This formulation is important because different actions require different kinds of contact. Support interactions, such as sitting, lying, standing, or leaning, depend on stable contact between load-bearing body regions and supporting scene surfaces. Manipulation interactions, such as opening, pulling, pressing, or lifting, depend on localized contact between the hands and functional object parts. Many interactions combine both types. A person opening a drawer must stand on the floor while reaching the handle. A person lifting a bin must grasp the object while maintaining a stable stance. A person leaning over a sink may contact the floor with the feet, the counter with the hands or torso, and the faucet with one hand. Representing these interactions as structured body-scene contacts makes these dependencies explicit.

The body-side representation also matters. SMPL-X provides a detailed parametric human

mesh, but the mesh vertices themselves are not semantically meaningful unless they are grouped into interaction regions. This thesis therefore uses manually defined contact regions on the SMPL-X body surface corresponding to interaction parts such as hands, arms, feet, pelvis, back, and other task-relevant regions. These regions provide a bridge between high-level language and low-level geometry: an instruction may mention a hand, a seated posture, a leaning action, or a grasping motion, while the final synthesis must place specific surface regions of the body near specific surface regions of the scene.

The scene-side representation is equally important. For many tasks, detecting the object category or even the object part is insufficient. The same cabinet can support several different interactions: opening a drawer, reaching for an upper shelf, leaning on the counter, or standing beside it. Each task selects a different scene region and a different set of body regions. Conversely, the same body part can participate in different functional roles: a hand can grasp a handle, press a switch, support the body on a tabletop, or hold the side of a bin. Interaction synthesis must therefore resolve not only which object is involved, but which object region is relevant and what type of body contact it requires.

Moreover, part-level grounding alone does not fully determine the interaction. Many object parts are spatially extended, and the exact location of contact within a detected part can strongly affect the plausibility of the synthesized pose. For example, when lifting a table, it is not enough to identify the tabletop as the relevant part; the hands should contact plausible grasp or support locations on the tabletop, often near accessible edges or corners rather than at arbitrary points on the surface. Similarly, for an interaction involving a ladder, the system must reason not only about the ladder as an object, or about its rails and rungs as parts, but also about which specific rail and rung regions should be contacted by the hands and feet. The same rung may support a foot at one location while a hand grasps a nearby rail at another, and these local choices determine whether the resulting pose is physically and functionally coherent.

This distinction is important because semantic part labels provide structure, but they do not by themselves specify contact geometry. A tabletop, rail, rung, seat, counter, cabinet front, or floor region can contain many possible contact locations, only some of which are compatible with the requested action, the human pose, the object affordance, and the surrounding scene. Body-part-scene-part grounding therefore requires a finer notion of localization: the framework must identify not only the relevant object and object part, but also the appropriate contact region within that part. This finer grounding allows the synthesized body to interact with the scene in a way that is both semantically correct and geometrically plausible.

To capture this structure, the thesis uses a Scene Interaction Graph as an intermediate representation. The graph expresses the target scene element, the participating human interaction parts, and the expected contact or support relations between them. Its purpose is not to describe every detail of the final pose, but to specify the interaction constraints

that the pose should satisfy. In this sense, the graph acts as a compact bridge between open-language task descriptions and 3D body synthesis. It turns an instruction such as “lift the trash bin with both hands” into a structured interaction hypothesis involving the bin, the two hands, the arms, the feet, and the supporting floor.

A key aspect of this thesis is that the scene-side contact regions are not assumed to be given by manual annotations, object geometry alone, or part labels alone. Instead, they are inferred through image-based generative reasoning. Starting from the scene image, a diffusion-based inpainting model first synthesizes a plausible human performing the requested interaction in the scene. This generated human provides a visual hypothesis about pose, orientation, laterality, and approximate contact layout. Then, conditioned on the interaction parts specified by the Scene Interaction Graph, a second diffusion-based inpainting step marks the corresponding contact regions on the target object or support surface in the original scene image. In this way, the scene image is used twice: first to imagine how the interaction should look, and then to localize where the relevant body parts should make contact on the object or scene.

This image-based contact grounding is particularly valuable for functional interactions, where the relevant contact region may be small, localized, or ambiguous even when the correct object part is known. A cabinet may contain many possible surfaces, but only the handle or drawer pull is appropriate for opening. A trash bin may have several visible faces, but only some are plausible grasp regions for lifting. Likewise, a large tabletop, ladder rail, or ladder rung may be semantically correct at the part level, but the interaction still depends on where within that part the hands, feet, or other body regions should make contact. By using diffusion models to propose both the human interaction hypothesis and the object-side contact regions, the framework leverages broad visual priors while keeping the final interaction tied to the actual scene and to the fine-grained spatial layout of the interaction.

Body-part–scene-part grounding is therefore the central abstraction behind this thesis. It makes the interaction explicit enough to guide synthesis, but flexible enough to support open-vocabulary instructions and previously unseen scenes. By grounding actions in relations between human surface regions and scene surface regions, and by further localizing contact within those scene regions, the formulation moves beyond object-level and part-level semantics toward a representation of how the body physically and functionally engages with the environment.

1.5 Problem Setting and Contributions

This thesis studies zero-shot static 3D human-scene interaction synthesis in real reconstructed indoor scenes. The input consists of a natural-language instruction, a calibrated

RGB image of the scene, and a reconstructed scene mesh. The output is a single SMPL-X body placed in the original scene coordinate frame. The synthesized body should express the requested action, contact the appropriate scene regions with the appropriate body parts, and remain physically plausible with respect to scale, orientation, support, and collision.

The setting is zero-shot because the method is not trained on paired 3D examples of the requested interaction in the target scene. The instruction may describe common support interactions, such as sitting on a chair, lying on a bed, or leaning on a table, or functional interactions, such as opening a drawer, pressing a switch, turning a faucet, or lifting a bin. In each case, the system must infer the relevant scene element, the functional or supporting region on that element, the participating human body parts, and the spatial arrangement required to realize the interaction. The target object and the required contact regions are not assumed to be available as ground-truth annotations.

The formulation is static. The goal is not to synthesize a full motion sequence, simulate forces, or modify the state of the scene. Instead, the output represents a single interaction state. For steady actions such as sitting or lying, this state corresponds to the interaction itself. For actions involving movable or articulated objects, such as opening a drawer, pulling a door, or lifting a trash bin, the synthesized body represents the pre-action or contact-ready state: the human has reached the relevant scene region and adopted a plausible pose, but the object has not yet moved. This choice isolates a fundamental subproblem of human-scene interaction: determining where contact should occur and how the body should be configured in a real 3D environment.

The proposed approach is built around body-part-scene-part grounding. An open-language instruction is first interpreted as a structured interaction involving human interaction regions, scene regions, and contact relations. On the human side, SMPL-X is treated not only as a parametric body model, but as a surface with semantically meaningful interaction regions, such as hands, arms, feet, pelvis, thighs, back, and torso. On the scene side, the relevant regions may be broad support surfaces or localized functional elements. The synthesis problem is then to place the body so that these body regions and scene regions are brought into physically and semantically meaningful correspondence.

To support this grounding in open-vocabulary scenes, the thesis combines structured interaction reasoning with generative visual priors and scene-aware 3D fitting. A Scene Interaction Graph represents the expected relations among the action, target scene element, participating body parts, and contact regions. Diffusion-based image reasoning provides visual hypotheses about how the interaction should appear and where the corresponding object-side contacts should lie. These hypotheses are then grounded in the reconstructed scene to synthesize a metric SMPL-X body that satisfies the inferred interaction structure. At a high level, the method uses foundation models to reason about what interaction is plausible, while using the scene reconstruction to ensure that the final body is placed in

3D rather than merely depicted in an image.

This thesis makes the following contributions:

1. **A zero-shot formulation of static 3D human-scene interaction synthesis.** The thesis defines the task of synthesizing a single SMPL-X body in a real reconstructed indoor scene from an open-language instruction, a calibrated RGB image, and a scene mesh, without requiring paired 3D interaction examples for the target task.
2. **A body-part–scene-part grounding view of HSI.** The thesis formulates interaction synthesis as the problem of grounding relations between human interaction regions and scene interaction regions, rather than only matching whole-body poses to object categories.
3. **A Scene Interaction Graph representation for open-vocabulary interactions.** The thesis introduces a structured representation that captures the target scene element, participating human body parts, scene regions, and expected contact or support relations implied by a natural-language instruction.
4. **A SMPL-X interaction-part abstraction for contact-aware synthesis.** The thesis defines semantically meaningful contact regions on the SMPL-X body surface, enabling high-level interaction descriptions involving hands, arms, feet, pelvis, back, and other body regions to be connected to low-level 3D body geometry.
5. **An image-guided strategy for localizing scene-side contact regions.** The thesis uses generative image reasoning to infer where the human interaction parts should contact the target object or support surface, allowing functional regions such as handles, knobs, drawer pulls, and graspable object areas to be identified without manual scene-side contact annotation.
6. **A scene-grounded synthesis framework for physically plausible body placement.** The thesis synthesizes a full SMPL-X body in the reconstructed scene coordinate frame, using the inferred interaction structure and scene geometry to encourage semantic consistency, appropriate contact, realistic scale, and reduced physical artifacts such as floating or penetration.

Together, these contributions move static HSI synthesis beyond object-level placement and toward interaction-level grounding. The thesis argues that plausible human-scene interaction requires knowing not only what object is involved, but which parts of the human body should engage which parts of the scene, and how that engagement can be realized as a coherent 3D body configuration.

1.6 Thesis Structure

Chapter 2 reviews the foundations and related work necessary for this thesis, including 3D scene reconstruction, parametric human body models, human-scene interaction synthesis, affordance and contact reasoning, foundation models for visual generation, and evaluation of physically grounded human-scene configurations.

Chapter 3 presents the proposed zero-shot static HSI synthesis framework. It describes the Scene Interaction Graph, the SMPL-X interaction-part abstraction, diffusion-based human and contact-region reasoning, scene-side contact grounding, and the final scene-aware SMPL-X fitting procedure.

Chapter 4 defines the experimental setup, including the ScanNet++ scenes, interaction instructions, selected action categories, implementation details, baselines or ablations, and quantitative and qualitative evaluation metrics.

Chapter 5 reports the experimental results and analyzes the behavior of the method. It examines semantic consistency, contact accuracy, physical plausibility, pose diversity, qualitative examples, ablation studies, and common failure modes.

Chapter 6 summarizes the findings of the thesis and discusses limitations and future directions, including dynamic interaction synthesis, richer physical reasoning, multi-view grounding, multi-human interactions, and more complete modeling of object state changes.

Chapter 2

Related Work

This chapter positions the thesis within prior work on human-scene interaction synthesis. It organizes the literature along two axes that are central to the thesis: trained versus zero-shot methods, and static 3D versus dynamic 4D interaction synthesis.

2.1 Foundations for Metric Human-Scene Interaction

Human-scene interaction synthesis requires reasoning jointly about the geometry of the environment, the articulation of the human body, and the physical contacts that make an interaction plausible. Unlike image-only human generation, where the output only needs to appear visually consistent from a particular viewpoint, 3D human-scene interaction requires the synthesized body to be metrically compatible with the surrounding scene. A body that looks plausible in a rendered image may still float above the floor, penetrate furniture, miss the intended contact surface, or touch an object part that is semantically irrelevant to the requested interaction. For this reason, the foundations of this thesis are metric scene reconstruction, parametric human body modeling, and explicit contact-aware optimization.

The scene side of the problem is represented by a reconstructed indoor environment together with calibrated image observations. Datasets such as ScanNet and ScanNet++ provide RGB-D views, camera parameters, and reconstructed 3D geometry, enabling image-level evidence to be related back to a common scene coordinate frame [1, 2]. This calibration is essential for the setting considered in this thesis. The method starts from a scene image and its camera parameters, but the desired output is not merely a 2D depiction of a person. Instead, the goal is to synthesize a 3D body that is placed in the original reconstructed scene. The availability of the camera intrinsics, extrinsics, and scene mesh makes it possible to project image-space cues into 3D, reason about body-scene distances, and enforce geometric constraints during optimization.

On the human side, this thesis uses SMPL-X as the body representation [3]. SMPL-X represents the human body as a differentiable parametric mesh with articulated body pose, hand pose, facial expression, shape, and global transformation parameters. Its differentiability makes it suitable for optimization-based refinement, while its dense surface provides a natural representation for contact reasoning. However, the SMPL-X mesh alone does not define which parts of the body should participate in an interaction. Physical and functional interaction reasoning requires additional semantic structure over the body surface. For example, grasping a handle primarily involves the inner hand surface, leaning may involve the back or shoulder region, sitting involves the pelvis, thighs, or hips, and stable standing requires foot-floor contact. Therefore, interaction synthesis benefits from grouping body vertices into contact-relevant regions rather than treating the body as an undifferentiated surface.

Previous work has shown that 3D human pose estimation and synthesis improve substantially when the surrounding scene is treated as an active constraint rather than as a passive background. PROX introduced scene-aware body fitting by extending monocular body estimation with contact and interpenetration constraints in reconstructed 3D scenes [4]. Its key observation is that many visually plausible monocular body estimates are physically inconsistent when inspected in 3D: the body may intersect furniture, miss supporting surfaces, or violate basic spatial constraints. PROX addresses this by penalizing body-scene penetration and encouraging selected body regions to contact nearby scene surfaces. This formulation is important for later HSI synthesis methods because it establishes contact and exclusion as explicit optimization objectives.

Subsequent work further developed contact as a learnable and semantically meaningful representation. POSA augments the SMPL-X body with body-surface contact probabilities and semantic scene labels, learning where the body is likely to contact the environment for plausible human-scene configurations [5]. PLACE similarly emphasizes proximity and contact between articulated humans and 3D environments [6]. These works demonstrate that realistic HSI cannot be reduced to estimating a natural human pose independently of the scene. Instead, plausible interaction depends on the compatibility between specific body regions and specific scene surfaces.

This body-region view of contact is particularly important for functional interaction. Many user instructions are not fully specified by object category alone. For example, opening a drawer requires contact with a drawer handle or front panel, not merely with the cabinet as a whole. Turning on a faucet requires reaching the control region, not simply touching the sink. Sitting, leaning, pushing, grasping, and stepping each imply different body-side contact regions and different scene-side contact surfaces. Thus, a central requirement for open-vocabulary HSI is to bridge high-level language descriptions and low-level metric contact constraints. The body must be articulated naturally, but it must also place the correct body part on the correct functional region of the scene.

Recent monocular human mesh recovery methods provide useful initialization for this optimization problem. In this thesis, GVHMR is used to obtain an initial SMPL-X body estimate from the inpainted human image [7]. GVHMR recovers world-grounded human motion by predicting human pose in gravity-view coordinates, which helps reduce ambiguity in monocular reconstruction. Although such methods can provide a strong initial body configuration, they do not by themselves guarantee correct interaction with a reconstructed scene. The recovered body may still require refinement to satisfy scene contact, avoid penetration, and align with the intended interaction semantics.

The final foundation is therefore optimization in the reconstructed scene. Given an initial SMPL-X body and projected scene-side contact targets, the body can be refined using losses that encourage contact with the target regions, penalize interpenetration with the scene mesh, and preserve a plausible human pose. Pose and orientation priors are important because contact objectives alone may produce unnatural configurations if the body is allowed to move freely. In this thesis, these priors also keep the optimized body close to the GVHMR initialization while allowing sufficient adjustment for physical interaction. This combination of image-based initialization, explicit body-part contact regions, projected scene contact targets, and scene-aware SMPL-X optimization provides the metric foundation for the proposed human-scene interaction synthesis pipeline.

2.2 Datasets and Supervision for HSI and HOI

Progress in human-scene interaction has been closely tied to the availability of datasets that capture scenes, bodies, objects, contacts, and language at different levels of detail. However, no single dataset provides all forms of supervision needed for open-vocabulary, fine-grained functional 3D HSI. Some datasets provide high-quality reconstructed scenes but no interacting humans; others capture humans in scenes but cover a limited number of environments and actions; object-centric datasets provide detailed human-object contacts but usually do not represent full reconstructed rooms; and language-conditioned motion datasets provide text supervision but often rely on synthetic alignments or closed action spaces. This diversity of datasets has shaped the design of existing methods and also motivates training-free approaches that use pretrained priors rather than requiring paired 3D human-scene interaction examples.

Early human-scene datasets established the importance of observing humans in metric environments. PiGraphs collected RGB-D observations of human activities and represented them as interaction graphs between body parts and scene geometry, linking human pose, object geometry, contact, and visual attention [8]. PROX later introduced RGB-D sequences of people interacting with reconstructed indoor scenes and showed that monocular body estimates can be improved by enforcing 3D scene constraints [4]. Such datasets are foundational because they make it possible to study contact, support, and collision in a

shared coordinate frame. At the same time, their scale and semantic coverage are limited: the number of scenes, object categories, and interaction types is small compared with the open-ended variety of everyday human-scene interactions.

A separate line of work focuses on high-quality static or reconstructed indoor scenes without necessarily providing paired human interaction data. ScanNet and ScanNet++ are examples of this category [1, 2]. They provide RGB-D or high-fidelity RGB observations, camera calibration, and reconstructed scene geometry, making them suitable for metric scene understanding. This thesis uses ScanNet++ as its scene source: each example consists of a scene image, the corresponding camera parameters, and the reconstructed scene mesh. This choice provides accurate geometry and camera alignment, but it also means that the method cannot rely on ground-truth humans, motions, or contact annotations in the target scenes. The task is therefore not to imitate captured interactions from the dataset, but to synthesize a plausible SMPL-X body in a previously reconstructed scene.

Human-object interaction datasets provide another important form of supervision. GRAB captures whole-body grasping motions with SMPL-X bodies, object poses, and contact between the body and manipulated objects [9]. BEHAVE provides multi-view RGB-D recordings, SMPL body fits, object fits, and annotated contacts for full-body human-object interactions [10]. InterCap further improves whole-body and object tracking from multi-view RGB-D input and includes interactions with daily objects where hand and foot contacts are important [11]. Other datasets such as COUCH, CHAIRS, and OMOMO focus on more specific interaction regimes, including chair interactions, articulated objects, and object-motion-guided full-body manipulation [12–14]. These datasets are valuable because they provide detailed evidence about how bodies contact objects. However, they are usually object-centric rather than full-scene and open-vocabulary. They do not directly solve the problem of locating the functional region of an arbitrary object in a reconstructed room from a natural language instruction.

Dynamic HSI datasets extend the supervision from isolated objects to motions in scenes. HUMANISE aligns human motions with 3D indoor scenes and adds language descriptions, enabling language-conditioned motion generation in 3D environments [15]. GIMO provides human motion, 3D scene context, egocentric views, and gaze information, supporting intent-aware motion prediction in real environments [16]. EgoBody captures 3D human pose and shape during social interactions from egocentric devices in complex scenes [17]. TRUMANS scales dynamic HSI further by capturing motion in many indoor scene configurations with contact-aware interactions [18]. These datasets support training models for navigation, sitting, standing, object-directed motion, and long-horizon interaction. Their main relevance to this thesis is that they illustrate the cost of paired motion-scene supervision: collecting accurate human motion, scene geometry, object states, and contact over time is expensive and difficult to scale to open-vocabulary tasks.

The type of supervision also determines the generalization behavior of trained HSI methods.

Methods trained on paired scene-motion data can learn strong priors for common actions, but they are constrained by the distribution of captured interactions. Methods trained with semantic action-object labels, such as COINS, can synthesize interactions from controlled action-object specifications, but the interaction vocabulary and object semantics remain predefined [19]. Methods trained with contact annotations can learn where bodies usually touch objects, but contact labels are expensive to obtain and often tied to a limited set of object categories or actions. These supervision regimes are effective when the test interaction resembles the training data, but they are less suitable when the instruction refers to an unseen functional element, an unusual scene layout, or a fine-grained object part such as a drawer handle, faucet control, cabinet edge, or window latch.

This thesis uses a different form of supervision. It does not train on paired 3D human-scene interaction data from ScanNet++ or any other HSI dataset. Instead, it uses pretrained models and explicit geometric optimization. The body-side contact regions are manually defined on SMPL-X as subsets of semantically meaningful parts, including hands, arms, back, head, legs, hips, and feet. These regions are selected to correspond to surfaces that commonly participate in contact, such as the inner hand surface, downward-facing foot vertices, and backside regions of the head, back, hips, and legs. These annotations are used both to define contact losses during optimization and to evaluate whether the synthesized interaction places the appropriate body region in contact with the scene. The scene-side contact regions, however, are not provided by the dataset. They are inferred from image-space evidence and projected back to the reconstructed mesh.

The resulting research gap is therefore not simply a lack of data, but a mismatch between available supervision and the desired task. Existing datasets provide realistic scenes, captured human motions, object-level contacts, semantic labels, or language descriptions, but they rarely provide open-vocabulary task descriptions paired with fine-grained functional contact regions on reconstructed scene meshes. This motivates a training-free formulation in which pretrained visual and generative priors propose interaction evidence, while explicit projection and SMPL-X optimization enforce metric consistency in the ScanNet++ scene coordinate frame.

2.3 Affordances, Functional Scene Elements, and Contact Representations

Human-scene interaction is governed not only by geometry but also by affordance: the possibility of action offered by a scene to a human body. In 3D HSI, affordance is usually expressed through spatial relations between the body and the environment, such as sitting on a support surface, standing on the floor, leaning against a wall, touching an object, or grasping a functional part. These relations are physical because they require metric

consistency, but they are also semantic because the same object can afford different actions depending on which part is used. A chair may support sitting on its seat, touching on its backrest, or stepping near its legs; a cabinet may afford leaning, touching, opening, or grasping depending on whether the relevant region is the side surface, front panel, handle, or drawer edge. Therefore, functional HSI requires more than detecting the target object: it requires identifying which body region should contact which scene region.

Early work on interaction-centered scene understanding made this connection explicit. PiGraphs represented human activities as probabilistic interaction graphs between human body parts and scene objects, showing that everyday actions can be described through recurrent geometric and semantic relations between people and their surroundings [8]. The key insight is that human pose provides evidence about scene function, and scene structure constrains plausible human behavior. This perspective remains important for later synthesis methods: a generated body should not simply be close to an object, but should instantiate a meaningful relation between an action, a body part, and a scene element.

Contact-aware body modeling further developed this idea by treating the body surface as a structured interaction representation. PROX introduced contact and interpenetration constraints for fitting SMPL-X bodies in reconstructed scenes, encouraging selected body regions to touch nearby scene surfaces while avoiding physically impossible intersections [4]. PLACE and POSA then used contact more directly for synthesis. PLACE models proximity and contact between articulated humans and 3D environments, while POSA learns body-surface contact probabilities and scene semantics on top of SMPL-X [5, 6]. These methods establish that plausible HSI depends on body-scene compatibility at the surface level, not only on global body pose or object proximity.

However, learned contact priors are often limited by the granularity of their supervision. A body-surface contact probability can indicate that the hand, foot, pelvis, or back is likely to contact the scene, but it does not necessarily specify the exact functional region on the target object. For many interactions, this distinction is crucial. In ‘sit on a chair,’ contacting the seat is more meaningful than touching a chair leg. In ‘turn on the faucet,’ the relevant region is the control or handle area, not the sink basin as a whole. In ‘take the bike off the stand,’ the hand contact should occur on a plausible graspable part of the bike, while the feet may remain supported by the floor. Thus, object-level affordance and body-level contact are both necessary but insufficient when considered separately.

Semantic-control methods address part of this limitation by conditioning HSI generation on higher-level interaction labels. COINS synthesizes human bodies in scenes from compositional action-object specifications, such as sitting on a chair or touching a table [19]. Its formulation is important because it recognizes interactions as semantic compositions rather than purely geometric arrangements. Nevertheless, the semantic unit remains an action-object pair. This is suitable for interactions where the object category strongly

determines the contact surface, but it is less precise for functional interactions where the relevant contact is on a small or ambiguous object subregion. An instruction such as ‘open the bottom drawer’ or ‘hold the side frame of the treadmill’ requires grounding below the object-instance level.

Dense contact estimation methods study contact more directly, often from images or motion. DECO estimates dense vertex-level human-scene contact on the body surface from a single RGB image, reasoning over body parts, proximity, and scene context [20]. SUMMON uses ContactFormer to infer temporally consistent contact labels from human motion and then synthesize plausible scenes around those contacts [21]. These works are relevant because they show that contact can serve as an intermediate representation between human motion, scene geometry, and interaction semantics. At the same time, they mostly predict or use contact on the human side, or infer object existence from motion, rather than localizing a task-specific contact region on an existing reconstructed scene mesh.

Recent zero-shot HSI methods move closer to open-vocabulary functional reasoning. GenZI uses vision-language and image generation priors to synthesize plausible 2D human-scene interaction hypotheses from rendered views, then optimizes a 3D human body in the scene [22]. FunHSI explicitly targets open-vocabulary functional 3D HSI by identifying functional scene elements, constructing a contact graph, synthesizing a human performing the task, and refining the 3D body for physical and functional correctness [23]. These works demonstrate that pretrained vision-language and generative models can provide interaction knowledge beyond closed action-object datasets. They also highlight a central challenge: foundation models can propose plausible interactions, but the final 3D output still requires explicit contact grounding and geometric verification.

This thesis builds on this contact-centered view but represents interaction in a deliberately simple and task-oriented form. The intermediate Scene Interaction Graph contains nodes for human body regions and the target object, or the floor when the interaction requires foot support. Edges specify which body regions should contact the target object or floor. The body regions are manually defined on SMPL-X as contact-specific subsets of broader anatomical parts, including hands, arms, back, head, legs, hips, and feet. For example, hand contact uses the inner hand surface, foot contact uses downward-facing foot vertices, and back-facing regions are selected for body parts such as the head, back, hips, and legs. This graph does not require assigning semantic names to scene subparts such as handle, knob, or edge. Instead, the graph identifies the required body-object contact relations, and the scene-side contact evidence is later inferred visually.

The floor is treated as a special case because foot support is often spatially flexible. For interactions such as standing, walking preparation, climbing, or manipulating an object while supported by the floor, the exact foot location may vary as long as the feet contact a valid floor region and the body remains physically plausible. In contrast, object contacts such as hands on a treadmill frame, hands on a ladder, or hips on a chair require more

localized scene-side evidence. This motivates separating generic support regions, such as floor contact, from target-object contacts that must be inferred from the visual and functional context of the instruction.

The main limitation of prior contact and affordance representations, from the perspective of this thesis, is therefore their mismatch with fine-grained open-vocabulary functional synthesis. Some methods learn likely human-scene contacts from paired data; others condition on object-level action labels; others estimate dense human-body contact from images or motion. Yet open-vocabulary functional HSI requires pairing the correct body contact region with the correct functional scene subregion in a specific reconstructed scene. The thesis addresses this gap by using the Scene Interaction Graph to determine the required body-side contacts, using generative visual priors to infer the corresponding scene-side contact regions, and then optimizing SMPL-X in the reconstructed scene coordinate frame.

2.4 Static 3D Human-Scene Interaction Synthesis

Static 3D human-scene interaction synthesis aims to generate a single plausible human body configuration in a 3D environment. Unlike dynamic HSI, which must model trajectories, temporal coherence, and changing contacts over time, static HSI focuses on whether one body pose is semantically appropriate and physically compatible with the scene. This problem is already challenging because the generated body must satisfy several coupled constraints: it should be anatomically plausible, placed at a meaningful location, oriented toward the relevant object or scene region, supported by nearby geometry, and free of severe interpenetration. For this reason, early static HSI methods focused on learning where humans can plausibly be placed in scenes and how body pose relates to surrounding geometry.

Putting Humans in a Scene is one of the early learning-based approaches for predicting human affordances in 3D indoor environments [24]. Rather than merely detecting objects, it learns what human poses are afforded by the scene, such as sitting on a chair or standing on the floor. The method uses synthesized training data produced by combining semantic knowledge from 2D human poses with 3D geometric knowledge from indoor scenes. This work is important because it frames human placement as an affordance prediction problem: the scene is not only an obstacle field, but a structure that implies possible human configurations.

Generating 3D People in Scenes without People extends this idea by generating full SMPL-X bodies in empty 3D scenes [25]. The method first predicts semantically plausible body configurations conditioned on the scene and then refines them using scene constraints to improve physical feasibility. Its formulation makes explicit that static HSI requires both

semantic plausibility and physical consistency. A person sitting on a sofa must not only be near the sofa, but must have a pose and placement that create support contact while avoiding penetration. This two-stage pattern, in which a learned model proposes a body and a subsequent optimization or refinement step enforces geometry, remains influential in later work.

PLACE and POSA further emphasize contact-aware interaction synthesis. PLACE introduces proximity learning of articulation and contact in 3D environments, using local scene geometry to generate bodies that are compatible with surrounding surfaces [6]. POSA learns a body-centric representation of human-scene contact by predicting contact labels and scene semantics on the SMPL-X surface [5]. These methods are significant because they move beyond global placement and treat contact as part of the representation of interaction. They show that plausible static HSI depends on local compatibility between body surfaces and scene surfaces. However, their learned priors are shaped by available interaction data and scene semantics. As a result, they are strongest for interactions that resemble the distributions observed during training.

Semantic control is introduced more explicitly in COINS, which synthesizes human-scene interactions from action-object specifications such as sitting on a chair, touching a table, or composing multiple atomic interactions [19]. COINS represents a major step from generic scene population toward controllable HSI: the user can specify not only that a body should be placed in a scene, but also which action should be performed with which object instance. Its compositional formulation is useful because many interactions can be decomposed into simpler action-object relations. Nevertheless, the interaction is still specified at the level of predefined action and object categories. This makes the method less suited to open-vocabulary functional instructions in which the relevant contact region may be a small object part or an ambiguous functional surface rather than the object instance as a whole.

Recent diffusion-based models extend static interaction synthesis toward richer object and contact control. TriDi models 3D human-object interaction by jointly generating human, object, and interaction modalities with a three-way diffusion process [26]. It is relevant to static HSI because it treats contact maps and text descriptions as controllable interaction representations, rather than using body pose or object pose alone. Although TriDi is primarily object-centric rather than full-scene-centric, it reflects a broader trend: static interaction synthesis increasingly depends on intermediate contact representations that connect semantic intent to geometric constraints. At the same time, such methods remain data-driven and are typically trained on object-level interaction datasets such as GRAB and BEHAVE, which differ from open-vocabulary interaction synthesis in reconstructed indoor scenes.

Across these static synthesis methods, the field has progressed from plausible body placement, to contact-aware body-scene compatibility, to semantic and contact-conditioned

interaction generation. These developments provide important foundations for this thesis. They show that static HSI is feasible when methods have access to learned priors, annotated contacts, semantic object labels, or paired interaction data. However, they also reveal limitations that are central to the proposed problem setting. Many methods depend on closed action or object vocabularies, require paired human-scene or human-object interaction data, and ground interactions at the object level rather than at the level of functional contact regions. This becomes limiting for instructions such as turning on a faucet, holding the side frame of a treadmill, climbing a ladder, or taking a bike off a stand, where the correct interaction depends on a precise contact region in the scene.

The thesis belongs to static 3D HSI as a task: it synthesizes a single SMPL-X body configuration that realizes a requested interaction in a reconstructed scene. However, it does not train a static HSI generative model from paired 3D interaction data. Instead, it uses language reasoning to determine body-part-scene relations, image generation to obtain a plausible human hypothesis, visual contact annotation to infer scene-side contact evidence, and optimization to enforce contact, collision, and pose plausibility in the reconstructed scene coordinate frame. This positions the work as a static HSI method that keeps the geometric accountability of contact-aware synthesis while avoiding dependence on closed interaction datasets.

2.5 Language-Conditioned and Semantically Controlled HSI

Language provides a natural interface for controlling human-scene interaction synthesis. A user may specify a simple action such as ‘sit on the chair,’ but may also describe a more complex interaction such as ‘a person on the treadmill with both hands on the side frame, one foot on the treadmill belt and one foot in the air.’ Such instructions contain different types of information: the intended action, the target object, spatial relations between objects, body-part constraints, and sometimes detailed contact requirements. A language-conditioned HSI method must therefore solve more than text-to-pose generation. It must ground the instruction in the scene, identify the relevant object or region, infer the human pose or motion implied by the action, and ensure that the final body-scene relation is physically plausible.

Early semantic-control methods approached this problem through structured action-object labels rather than free-form language. COINS synthesizes compositional human-scene interactions from semantic specifications defined as action-object pairs, such as sitting on a chair or touching a table [19]. This formulation is important because it introduces controllability into static HSI: the generated body is conditioned not only on the scene geometry but also on the intended interaction. However, the language interface is limited by a predefined set of actions and object categories. The method can compose known action-object relations, but it does not directly address open-vocabulary descriptions,

fine-grained object parts, or instructions where the functional contact region must be inferred from visual context.

Language-conditioned motion generation extends semantic control to temporal HSI. HUMANISE introduced language-conditioned human motion generation in 3D scenes, connecting text descriptions with scene-aware motion sequences [15]. It showed that text can specify both human action and spatial intent in a scene, such as walking to or interacting with a target object. However, grounding language in a 3D scene is difficult when the entire scene is encoded as a large geometric input. The model must implicitly learn which object the language refers to and how the generated body should relate to it. This motivates later methods that decompose language-conditioned HSI into more explicit grounding and generation stages.

TextSceneMotion addresses this issue by separating target-object grounding from object-centric motion generation [27]. Instead of generating motion directly from the whole scene and text, it first grounds the object described by the language and then generates motion relative to that object. This decomposition is conceptually important for this thesis because it treats language grounding as a distinct problem rather than expecting a generative model to infer all spatial correspondences implicitly. The limitation is that the grounding target is still primarily the object instance, whereas functional HSI often requires a smaller contact region on the object surface.

TeSMo also studies text-controlled scene-aware motion generation, using diffusion models to generate motions such as walking, sitting, and interacting with objects in 3D scenes [28]. It combines text control with start and goal constraints, allowing the user to guide both motion style and spatial intent. Move as You Say, Interact as You Can further introduces scene affordance maps as an intermediate representation between language grounding and motion generation [29]. Its two-stage structure first predicts an explicit affordance map and then generates motion conditioned on that affordance representation. This is relevant because it shows that intermediate representations can help bridge the gap between free-form language and physically grounded motion. In this thesis, the intermediate representation is not an affordance map for motion generation, but a Scene Interaction Graph that specifies body-part-object or body-part-floor contact relations for static synthesis.

Open-vocabulary HSI methods use foundation models to reduce dependence on closed semantic vocabularies. GenZI generates zero-shot 3D human-scene interactions from natural language descriptions and 3D scenes by using vision-language and image inpainting models to synthesize plausible 2D interaction hypotheses, which are then lifted to a 3D body through optimization [22]. This demonstrates that pretrained 2D generative models contain useful priors about how humans interact with scenes. However, the language input mainly guides the generation of plausible human-scene compositions; the method does not explicitly focus on extracting fine-grained scene-side contact regions for specific body parts.

FunHSI is more directly aligned with functional language-conditioned interaction. It takes open-vocabulary task prompts and reasons about functional scene elements, contact graphs, and 3D body refinement [23]. This is an important step beyond object-category grounding because it recognizes that instructions such as increasing room temperature or opening a window require reasoning about object functionality. FunHSI shows that functional HSI benefits from explicitly representing which parts of the scene are relevant to the task and which contacts make the interaction correct. The present thesis follows the same broad motivation, but uses a different route to contact grounding: language is first converted into a Scene Interaction Graph over body parts and the target object or floor, while the precise scene-side contact region is inferred visually from generated contact annotations.

Language-conditioned HSI also appears in video generation. GenHSI decomposes controllable HSI video generation into script writing, 3D-aware keyframe generation, and animation with a pretrained video diffusion model [30]. This pipeline is relevant because it treats language as a high-level controller that can be decomposed into atomic interaction steps before visual synthesis. It also illustrates the increasing role of foundation models in interpreting interaction descriptions and producing plausible human-scene configurations. Nevertheless, video-oriented methods emphasize temporal coherence and visual plausibility, while the static 3D setting in this thesis emphasizes metric contact, mesh alignment, and body-region-specific optimization.

The key distinction in this thesis is the granularity at which language is grounded. Prior language-conditioned HSI methods commonly map language to action labels, target objects, affordance maps, or motion trajectories. In contrast, this thesis maps language to explicit body-part–scene contact requirements. The Scene Interaction Graph specifies which human regions, such as hands, arms, back, head, hips, legs, or feet, should contact the target object or floor. The method does not require predefined semantic names for object parts such as handle, knob, edge, or seat. Instead, after language reasoning determines the body-object contact structure, visual generative models infer where those contacts should occur on the scene image, and the resulting evidence is projected to the reconstructed scene for optimization. This allows both short prompts and detailed compositional prompts to be handled within the same contact-grounded static HSI formulation.

2.6 Scene-Aware Human Motion and 4D HSI

Dynamic human-scene interaction extends static body placement to motion over time. Instead of synthesizing one plausible configuration, a 4D HSI method must generate a temporally coherent sequence in which the body navigates through the environment, avoids obstacles, makes and breaks contacts, and satisfies the intended action. This introduces constraints that do not appear in single-frame synthesis: foot motion should not slide, trajectories should remain feasible, transitions should be smooth, contacts should be

temporally consistent, and object interactions may involve changing object state. As a result, scene-aware motion synthesis has developed a distinct set of representations and evaluation criteria, including trajectory realism, collision avoidance, foot contact, temporal smoothness, long-horizon consistency, and semantic alignment with an action or text instruction.

Early scene-aware motion methods focused on generating long-term human behavior in 3D environments by combining motion priors with scene constraints. Wang et al. proposed a hierarchical framework for synthesizing long-term 3D human motion conditioned on scene structure, using geometry constraints between the human mesh and scene point clouds to improve physical plausibility [31]. Scene-aware generative models later factorized motion into trajectory generation and pose dynamics conditioned on the surrounding scene [32]. These works helped establish that motion generation cannot be treated as scene-agnostic animation followed by placement: the environment determines where the body can move, where it can stand, and which objects can support or constrain an action.

Motion-primitive and policy-based methods further introduced controllability and long-horizon synthesis. SAMP models stochastic scene-aware motion prediction, allowing virtual humans to navigate complex scenes and perform goal-directed interactions with target objects [33]. GAMMA generates long-term motion using generative motion primitives represented by body-surface markers, enabling diverse digital humans to move through 3D scenes while maintaining plausible body-ground contact [34]. DIMOS formulates human behavior synthesis in 3D indoor scenes as a policy-learning problem with latent motion action spaces, combining navigation, collision avoidance, and interaction goals to produce diverse motions such as wandering, sitting, lying, and standing up [35]. These methods are important because they connect scene-aware interaction to planning and control, but they still require learned motion priors and are primarily designed for whole-body motion over time rather than fine-grained static contact localization.

Diffusion models have become a dominant paradigm for scene-aware motion generation. SceneDiffuser formulates scene-conditioned generation, optimization, and planning as a unified diffusion process that can handle human pose and motion generation, dexterous grasp generation, path planning, and robot motion planning in 3D scenes [36]. This is significant because diffusion sampling can combine generative diversity with constraint-driven refinement. TeSMo uses denoising diffusion for text-controlled scene-aware motion generation, first learning a scene-agnostic text-to-motion model and then introducing scene-aware components for navigation and interaction [28]. HUMANISE and TextSceneMotion similarly use language to control human motion in 3D scenes, with TextSceneMotion explicitly decomposing the problem into target-object grounding and object-centric motion generation [15, 27]. Together, these methods show that language, scene geometry, and motion priors can be integrated, but their output is a motion sequence rather than a single optimized contact state.

A key design question in dynamic HSI is how to represent the scene. Dense 3D representations such as point clouds, voxel grids, signed distance fields, and occupancy grids preserve geometric information but can be expensive and difficult to scale. Object-centric representations reduce scene complexity but may lose global layout context. Recent work explores hybrid or lightweight alternatives. SceMoS separates global planning from local physical execution by using a bird’s-eye-view image with DINOv2 features for high-level scene reasoning and local 2D heightmaps for geometry-grounded motion tokenization [37]. SceneMI represents scene-aware motion in-betweening with global scene descriptors and local body-centered Basis Point Set features, using diffusion to synthesize transitions that satisfy keyframe constraints while adapting to environmental affordances [38]. These methods are relevant because they illustrate that dynamic HSI often requires both global layout understanding and local contact-aware geometry.

The scale of supervision is another major issue. Dynamic HSI benefits from paired motion-scene datasets because the model must learn not only plausible static contacts but also how humans approach, use, and leave objects over time. TRUMANS directly addresses this need by collecting a large motion-captured HSI dataset with diverse interactions across many indoor scene configurations and by training a diffusion-based autoregressive model for long, scene-conditioned interaction sequences [18]. The dataset and model demonstrate how performance improves when large-scale scene-motion supervision is available. At the same time, the need for such data highlights a limitation for open-vocabulary HSI: collecting exhaustive motion, object-state, and contact supervision for arbitrary scene functions remains difficult.

Zero-shot 4D HSI methods attempt to reduce this dependence by using video generation models as dynamic interaction priors. ZeroHSI distills human-scene interactions from video generation models, generating a video of a human interacting with a 3D scene and then reconstructing the 4D interaction through differentiable rendering [39]. Its motivation is closely related to zero-shot static HSI: pretrained visual models have observed a broad range of human interactions and can provide priors that are difficult to capture in paired 3D datasets. GenHSI also uses foundation video and image models, but decomposes controllable HSI video generation into script writing, 3D-aware keyframe generation, and animation [30]. By generating keyframes that encode important interaction states before using video diffusion for temporal interpolation, GenHSI shows that static interaction poses can serve as anchors for dynamic HSI.

Despite these advances, dynamic HSI methods solve a different problem from this thesis. Their central challenge is to generate plausible motion through time: trajectories, transitions, foot contact, temporal smoothness, and long-horizon consistency. The thesis instead focuses on a static but more geometrically accountable target: a single SMPL-X body configuration that realizes an open-vocabulary interaction in a reconstructed ScanNet++ scene. This narrower output allows the method to emphasize body-region–scene-region

contact correctness, collision avoidance, and metric alignment. In particular, it allows the scene-side contact region to be inferred from visual generative evidence and then enforced directly during SMPL-X optimization, rather than being treated as an implicit property of a motion sequence.

2.7 Foundation Models for Zero-Shot HSI

Recent progress in foundation models has changed the design space for human-scene interaction synthesis. Instead of learning every interaction prior from paired 3D human-scene data, zero-shot methods use pretrained models as sources of semantic, visual, geometric, and motion knowledge. Large language models can decompose instructions and infer likely interaction structure; vision-language models can judge whether a rendered or generated image matches an action; image diffusion models can synthesize plausible human-scene compositions; video diffusion models can provide priors over temporal interaction; segmentation models can localize objects and regions in images; and monocular human reconstruction models can initialize parametric bodies. These models do not directly solve metric 3D HSI, but they provide hypotheses that can be grounded through camera calibration, scene geometry, and optimization.

Image diffusion models are particularly important because they encode broad visual priors over people, objects, and everyday interactions. Latent diffusion models showed that high-quality image synthesis and inpainting can be performed efficiently in a compressed latent space [40]. ControlNet further demonstrated that pretrained text-to-image diffusion models can be guided by spatial controls such as depth, edges, segmentation, and human pose [41]. For HSI, this means that 2D generative models can be used not only to synthesize a person in a scene, but also to provide visual evidence about pose, orientation, object use, and plausible contact. However, such evidence is image-space and viewpoint-dependent. To be useful for 3D HSI, it must be converted into metric constraints in the reconstructed scene.

GenZI is a key example of this strategy in zero-shot static 3D HSI [22]. Given a 3D scene, a text prompt, and a coarse interaction location, GenZI renders the scene from multiple views and uses vision-language and image inpainting models to imagine plausible 2D human-scene interactions. It then optimizes a 3D human body to match these generated 2D hypotheses. This shows that 2D foundation models can provide interaction priors without training on paired 3D HSI data. The important limitation, from the perspective of this thesis, is that the generated image primarily acts as an implicit pose and appearance hypothesis. Fine-grained scene-side contact regions, such as a handle, ladder rung, treadmill frame, or drawer edge, are not treated as explicit optimization targets.

FunHSI moves zero-shot HSI closer to functional reasoning [23]. It addresses open-

vocabulary task prompts by identifying functional scene elements, constructing a contact graph, synthesizing a human performing the task, and refining the 3D body through optimization. This is closely related to the motivation of this thesis because many interactions cannot be solved from object category alone. Instructions such as ‘increase the room temperature,’ ‘open the bottom drawer,’ or ‘turn on the faucet’ require reasoning about object functionality and the body configuration needed to activate that functionality. FunHSI demonstrates the value of combining foundation-model reasoning with explicit contact structure. The present thesis follows this broad direction, but uses a different mechanism for scene-side grounding: it does not require assigning semantic names to object parts. Instead, the precise contact region is inferred visually as colored contact annotations on the scene image and then projected to the 3D mesh.

PhySIC is related from the reconstruction side rather than the synthesis side [42]. It reconstructs physically plausible human-scene pairs and dense contact from a single RGB image by combining monocular cues with optimization over body pose, scene geometry, global scale, contact, and interpenetration. Although PhySIC assumes an observed human rather than generating a new human from an instruction, it is relevant because it illustrates the same principle that appears in zero-shot HSI: pretrained perceptual estimates are insufficient unless they are reconciled with physical constraints. In both reconstruction and synthesis, the final result must be evaluated in 3D, not only in the original image.

Foundation models have also influenced human-object interaction generation. HOI-Diff synthesizes text-driven 3D human-object interactions using diffusion models, including mechanisms for modeling contact between the human and object [43]. InterDiff uses physics-informed diffusion to anticipate future human-object interactions with dynamic objects [44]. DreamHOI uses text-to-image diffusion priors for zero-shot 3D human-object interaction synthesis with a given object and textual description [45]. InterFusion similarly uses text and diffusion-based visual priors for 3D human-object interaction generation [46]. HOI-PAGE is especially relevant because it introduces Part Affordance Graphs for zero-shot 4D HOI generation, using large language models to reason about how human body parts should engage with object parts before generating reference interaction videos and optimizing part-level contact constraints [47]. These works show that diffusion and language priors can help bridge high-level descriptions and physical interaction. However, object-centric HOI differs from full-scene HSI: the target object is usually isolated or explicitly provided, whereas HSI requires reasoning about an object within a larger scene, resolving spatial context, floor support, occlusion, surrounding obstacles, and contact with the reconstructed environment.

Video foundation models extend these ideas to 4D HSI. ZeroHSI uses video generation models as dynamic interaction priors and reconstructs human-scene interaction motion through differentiable rendering [39]. This is significant because it treats pretrained video generation not merely as a visual output mechanism, but as a source of interaction

knowledge learned from large-scale video data. GenHSI uses a training-free pipeline that decomposes HSI video generation into script writing, 3D-aware keyframe generation, and animation with a pretrained video diffusion model [30]. In both cases, foundation models provide plausible temporal interaction hypotheses. For this thesis, these methods are relevant because they show that generated images or videos can serve as intermediate representations for 3D interaction synthesis. The difference is that this thesis uses a static image-based formulation to extract and enforce explicit contact regions rather than reconstructing full motion.

Segmentation foundation models are another important component for grounding visual hypotheses. Segment Anything introduced promptable image segmentation and showed strong zero-shot transfer to new image distributions and tasks [48]. SAM 3 extends this direction toward concept-prompted segmentation in images and videos, returning masks and instance identities for objects specified by noun phrases, exemplars, or combinations of prompts [49]. Such models are useful in HSI because contact reasoning often requires separating the target object, floor, or support surface from the rest of the scene image. In this thesis, floor contact is treated as a special case: when the Scene Interaction Graph includes foot-floor contact, floor segmentation in the view is obtained and projected to the scene mesh. This differs from target-object contact, where the precise functional region is inferred from visual contact annotations rather than from a generic object mask.

Pretrained human mesh recovery models provide another form of foundation prior. Monocular human reconstruction methods can estimate a parametric body from an image, giving an initial pose, shape, and global configuration that can be refined in the scene. This thesis uses GVHMR to initialize the SMPL-X body from the inpainted human image [7]. GVHMR is useful because it predicts a world-grounded human configuration rather than only an image-aligned body estimate. Nevertheless, the recovered pose is still only an initialization. It may not satisfy the intended contact relation or avoid all scene collisions. Therefore, the pose must be refined using contact, penetration, pose-prior, and orientation-prior losses.

The proposed pipeline combines these foundation-model roles into a zero-shot static HSI system. A language model converts the user instruction into a Scene Interaction Graph, whose nodes are human body regions and the target object or floor, and whose edges specify required contacts. A detailed prompt is used to inpaint a plausible human into the ScanNet++ scene image. GVHMR estimates an initial SMPL-X body from this inpainted human image. The inpainted human image and original scene image are then used to ask a generative image model to mark each required scene-side contact region with a distinct bright color, for example using different colors for left-hand and right-hand contacts. These colored overlays are converted into binary masks and projected back to the reconstructed scene mesh. Finally, SMPL-X optimization enforces the projected contact targets while penalizing interpenetration and keeping the optimized body close to the initialization.

The common limitation of foundation-model-based HSI is that foundation models operate

primarily in language, image, or video space. They can propose plausible interactions, infer likely body-object relations, or hallucinate visually convincing contacts, but they do not automatically provide metrically valid 3D constraints. For a reconstructed-scene output, their predictions must be grounded in camera geometry, scene meshes, and physically meaningful optimization objectives. This thesis addresses that conversion explicitly: foundation models provide semantic and visual interaction priors, while projection and SMPL-X optimization convert those priors into contact targets and a final body configuration in the reconstructed 3D scene.

2.8 Image Diffusion, Segmentation, and 2D-to-3D Contact Localization

Image-space foundation models provide a powerful source of visual interaction priors for human-scene interaction synthesis. Text-to-image and image-editing diffusion models have learned broad regularities about how people appear in scenes, how bodies are posed relative to objects, and which visual configurations are plausible for everyday actions. In HSI, these priors are useful because many interactions are easier to imagine in an image than to specify directly as 3D constraints. However, image generation alone is insufficient for 3D synthesis. A generated person may look plausible from the input view while still being geometrically inconsistent with the reconstructed scene. The central challenge is therefore to convert image-space visual evidence into 3D constraints that can guide body optimization.

Diffusion-based inpainting and editing provide the basis for using generated images as interaction hypotheses. Latent diffusion models enable high-quality image synthesis and inpainting in a compressed latent space [40], while methods such as RePaint and InstructPix2Pix demonstrate the effectiveness of diffusion models for image completion and instruction-guided editing [50, 51]. ControlNet further shows that pretrained image diffusion models can be guided by spatial conditions such as depth, edges, segmentation, and pose [41]. More recent instruction-following image generation systems, including GPT-Image and Nano Banana 2, extend this capability with stronger multimodal prompting and image editing behavior [52, 53]. These models make it possible to ask for a human to be inserted into a given scene image according to a natural language instruction, producing a visually plausible interaction hypothesis without training a task-specific HSI model.

Several HSI and HOI methods already use 2D generative priors as bridges to 3D interaction. GenZI renders a 3D scene into image views, uses image inpainting and vision-language reasoning to synthesize plausible 2D human-scene interactions, and then lifts these hypotheses to a 3D human body through optimization [22]. GenHSI similarly uses image generation to create 3D-aware keyframes before video diffusion produces continuous HSI videos [30]. DreamHOI and related zero-shot HOI methods use text-to-image diffusion priors to syn-

thesize human-object interaction cues for 3D interaction generation [45]. HOI-PAGE is especially relevant because it uses part-level affordance reasoning to guide zero-shot 4D human-object interaction generation, connecting human body parts and object parts before generating interaction videos and optimizing contacts [47]. These works show that visual generative models can provide interaction hypotheses beyond what is available in paired 3D datasets.

The limitation is that most generated visual hypotheses remain implicit. A generated image may show a hand near a handle, a body sitting on a chair, or a foot on a ladder rung, but the image itself does not directly define which scene mesh vertices should be contacted by which SMPL-X body region. For metric HSI, this distinction is important. The output must not only look like an interaction from the camera view; it must also satisfy contact and collision constraints in 3D. Thus, a useful image prior should ideally be converted into explicit scene-side contact targets rather than used only as a qualitative visual guide.

Segmentation and visual localization models provide one way to convert image evidence into spatial constraints. Segment Anything introduced promptable image segmentation for zero-shot mask extraction [48]. SAM 3 extends this direction with concept-prompted segmentation in images and videos, making it possible to segment regions specified by semantic prompts [49]. In this thesis, segmentation is used in a targeted way rather than as a general object grounding pipeline. Floor contact is treated as a special support case: when the Scene Interaction Graph contains foot-floor edges, SAM 3 is used to obtain a floor mask in the input view, which can then be projected to the scene mesh. This is appropriate because the exact foot location on the floor is often flexible, provided that the feet remain supported and physically plausible.

Target-object contacts require a different treatment. For many interactions, the contact region is not the whole object mask and may not have a stable semantic label. A hand may need to contact a treadmill side frame, a faucet control, a bike handle, a ladder rung, a drawer front, or an edge of an object. Instead of first detecting and naming such object parts, this thesis uses the interaction image itself as evidence for where contact should occur. After a human is inpainted into the ScanNet++ view according to the detailed interaction prompt, the inpainted human image and the original scene image are given to an image generation model together with a Scene Interaction Graph-based instruction. The model is asked to mark each required scene-side contact region on the original scene image using a distinct bright color, such as red for one hand contact and blue for another. The colored overlays are then converted into binary masks.

This formulation differs from standard segmentation and image editing. The goal is not to segment the object category, edit the image for visual realism, or produce a final 2D result. Instead, the generated colored annotations act as an intermediate contact representation. Because the marks are produced on the original scene image without the inpainted human, they can be associated with visible scene regions rather than with the generated body

appearance. These 2D masks are then projected back to the reconstructed ScanNet++ mesh using the calibrated camera and scene geometry. The projected regions become explicit scene-side contact targets for SMPL-X optimization.

This 2D-to-3D conversion is the key link between foundation-model reasoning and metric HSI. Image generation provides a visual prior about where the interaction should occur; segmentation provides support-region masks when the target is geometrically broad, such as the floor; projection transfers the image-space evidence to the scene mesh; and optimization enforces the resulting contact constraints on the SMPL-X body. This allows the method to use flexible open-vocabulary image models while retaining a 3D output that is accountable to contact, penetration, pose prior, and orientation prior losses.

The main gap addressed by this thesis is therefore not simply the use of image diffusion for HSI, but the use of image diffusion for contact localization. Prior work uses diffusion and segmentation primarily to generate plausible humans, edit images, create interaction videos, or segment objects. In contrast, this thesis uses image generation to annotate scene-side contact regions conditioned on body-part contact requirements. The generated colored overlays are not the final output; they are converted into 3D contact targets and used to optimize a static SMPL-X body in the reconstructed scene.

2.9 Geometry-Aware Optimization for Physical Plausibility

Geometry-aware optimization is a recurring component in human-scene interaction because generative or monocular predictions alone rarely satisfy metric scene constraints. A predicted or generated body may be visually plausible from a camera view but physically invalid in 3D, for example by floating above the floor, penetrating furniture, missing the intended contact surface, or rotating away from the target object. Optimization provides a way to correct these errors by refining a parametric body model with respect to the reconstructed scene. In this thesis, optimization is especially important because foundation models provide image-space hypotheses, while the final output must be a physically plausible SMPL-X body in the ScanNet++ scene coordinate frame.

The optimization tradition in human body recovery begins with fitting parametric body models to observations under pose and shape priors. SMPLify fits SMPL to 2D keypoints using an optimization objective that combines reprojection consistency with body priors and interpenetration penalties [54]. SMPLify-X extends this idea to SMPL-X, enabling joint fitting of body, hands, and face from monocular images [3]. These methods show the value of optimizing a differentiable body model rather than directly regressing a mesh. However, image-only fitting is underconstrained: many 3D body configurations can explain the same 2D evidence, and without the surrounding scene the recovered body can violate physical constraints.

PROX addresses this limitation by adding scene constraints to monocular SMPL-X fitting [4]. It introduces an interpenetration term that penalizes the body for occupying the same space as the reconstructed environment and a contact term that encourages plausible body regions to touch nearby scene surfaces. This is a crucial step for HSI because it treats the scene as part of the optimization problem rather than as a background for rendering. PROX shows that contact and collision constraints can reduce pose ambiguity and produce bodies that are more consistent with the physical world. Later HSI synthesis methods inherit this idea by using scene geometry not only to evaluate results but also to refine them.

PLACE and POSA further connect optimization with learned contact priors. PLACE uses local scene geometry and proximity-aware representations to generate bodies that are compatible with 3D environments, followed by refinement for physical plausibility [6]. POSA predicts body-surface contacts and scene semantics on SMPL-X, providing a learned prior for where contact should occur between the body and scene [5]. These methods demonstrate that optimization benefits from good contact proposals: the optimizer needs targets or priors that indicate which body regions should approach which scene surfaces. Without such guidance, contact losses can be ambiguous, especially in cluttered scenes where many surfaces are nearby but only one is functionally relevant.

Zero-shot HSI methods also rely on optimization to convert foundation-model hypotheses into 3D bodies. GenZI uses generated 2D interaction hypotheses from rendered views and optimizes a 3D human body to match them in the scene [22]. FunHSI refines proposed 3D body and hand poses through a stage-wise optimization process to improve physical plausibility and functional correctness [23]. PhySIC, although focused on reconstruction rather than synthesis, similarly combines monocular perceptual estimates with physical optimization over body pose, scene geometry, contact, interpenetration, and scale [42]. These works show a common pattern: foundation or perception models provide initialization and semantic cues, while optimization enforces geometric consistency.

Efficient collision and penetration modeling is an important part of such optimization. Traditional mesh-based collision handling can be expensive or unstable when many body and scene points must be tested. Recent neural volumetric body models provide more efficient alternatives. VolumetricSMPL extends SMPL-family models with volumetric signed-distance functionality, enabling fast and differentiable SDF queries for interactions, contacts, and collision detection [55]. This is useful for HSI optimization because scene points can be queried against the posed body volume to penalize points that lie inside the body. In this thesis, penetration is handled by sampling scene points and evaluating them with the SMPL-X SDF provided by VolumetricSMPL, allowing the optimization to discourage scene-body intersections while remaining compatible with the SMPL-X body parameterization.

The proposed optimization follows this general lineage but uses a different source of

contact targets. The initial body is obtained from GVHMR applied to the inpainted human image, providing SMPL-X translation, rotation, and pose parameters [7]. The optimization then refines only the body translation, global rotation, and pose. Shape, camera parameters, and scene geometry are kept fixed. Contact is enforced by minimizing distances between selected body-side contact vertices and projected scene-side contact vertices. Penetration is penalized using the VolumetricSMPL SDF. Pose and orientation priors are implemented as penalties that keep the optimized SMPL-X parameters close to the GVHMR initialization, preventing contact objectives from producing unnatural deformations or implausible rotations.

The key difference from prior optimization-based HSI is the origin and granularity of the contact constraints. Previous methods often use learned contact priors, observed human contacts, nearby surfaces, or object-level interaction labels. In contrast, this thesis uses diffusion-derived scene-side contact regions as explicit optimization targets. The contact regions are inferred in image space from the original scene image, the inpainted human image, and the Scene Interaction Graph, then projected to the reconstructed mesh. Optimization therefore acts as the final grounding step: it converts visually inferred contact evidence into a physically plausible SMPL-X configuration while preserving the monocular initialization and avoiding scene penetration.

2.10 Summary and Research Gap

Prior work on human-scene interaction has established several important foundations for this thesis. Static 3D HSI methods show that plausible human placement requires contact-aware body-scene compatibility, semantic control, and collision avoidance [5, 6, 19, 24, 25]. Dynamic and 4D HSI methods extend this setting to trajectories, long-horizon motion, temporal contact consistency, and text-conditioned scene-aware motion [15, 18, 27, 28, 35–37]. These works demonstrate that HSI can be modeled effectively when sufficient supervision, scene representations, and interaction priors are available. However, many of them depend on paired human-scene or human-object interaction data, predefined action spaces, object-level semantic labels, or motion datasets whose coverage is necessarily limited relative to open-vocabulary everyday interactions.

Zero-shot and foundation-model-based methods reduce this dependence by using pretrained language, vision-language, image generation, video generation, segmentation, and monocular reconstruction models as sources of interaction knowledge. GenZI shows that 2D visual priors can be distilled into 3D HSI without paired 3D interaction training data [22]. FunHSI extends this direction toward open-vocabulary functional interaction by reasoning about functional scene elements and contact structure [23]. ZeroHSI and GenHSI show that video and image generation models can serve as priors for dynamic HSI and interaction keyframes [30, 39]. Object-centric HOI methods, including HOI-Diff, DreamHOI, InterFusion, and

HOI-PAGE, further show that diffusion and language priors can support contact-aware human-object interaction generation, with HOI-PAGE being especially relevant for its part-level affordance reasoning [43, 45–47]. Nevertheless, these methods often focus on visual plausibility, object-level interaction, or motion generation rather than precise static body-region–scene-region contact grounding in a reconstructed full scene.

The key research gap addressed in this thesis is fine-grained functional contact grounding for zero-shot static 3D HSI. Open-vocabulary instructions frequently require contact with a specific scene subregion rather than with an entire object instance. A person turning on a faucet, holding a treadmill side frame, climbing a ladder, taking a bike off a stand, or sitting on a sofa while interacting with a nearby object must place the correct body region on the correct functional scene region. Existing approaches provide important ingredients: learned contact priors, action-object semantic control, language-conditioned motion generation, foundation-model visual priors, and optimization-based physical refinement. What remains underexplored is the direct conversion of generated visual interaction evidence into explicit 3D contact targets for body optimization.

This thesis introduces a zero-shot static HSI formulation for ScanNet++ scenes that combines open-vocabulary reasoning with diffusion-based fine-grained contact localization. Given a scene image, camera parameters, and reconstructed scene mesh, the method first uses language reasoning to determine which body regions should contact the target object or floor. An image generation model then inpaints a plausible human performing the requested interaction. The central novelty is that image generation is also used to produce colored contact annotations on the original scene image: each required contact region is marked visually according to the body-region contact requirements. These colored annotations are converted into masks, projected onto the reconstructed mesh, and used as explicit scene-side contact targets.

The final SMPL-X body is obtained by optimizing the GVHMR initialization in the reconstructed scene coordinate frame using contact, penetration, pose-prior, and orientation-prior losses. This connects foundation-model flexibility with metric geometric accountability. In contrast to methods that rely on closed interaction datasets or use generated images only as implicit visual hypotheses, the proposed approach treats diffusion-generated contact evidence as an intermediate representation for 3D optimization. In this way, the thesis builds on prior work in contact-aware HSI, language-conditioned interaction, zero-shot visual priors, and geometry-aware refinement, while focusing specifically on the missing problem of fine-grained functional contact grounding in static 3D human-scene interaction.

Chapter 3

Methodology

This chapter describes the proposed zero-shot static human-scene interaction synthesis pipeline. The method converts a natural-language instruction, a calibrated ScanNet++ RGB view, and a reconstructed scene mesh into a single SMPL-X body placed in the original scene coordinate frame.

3.1 Overview

The proposed method synthesizes a static 3D human-scene interaction from three inputs: a natural-language interaction instruction y , a calibrated RGB image I of the scene, and a reconstructed scene mesh M_s . The output is a single SMPL-X body, represented by optimized body parameters and exported meshes, aligned to the ScanNet++ scene coordinate frame [2, 3]. The output is static: it represents the interaction state or contact-ready pre-action state, not a full motion sequence and not an edited object state.

Figure 3.1 summarizes the pipeline. First, a vision-language model converts the instruction and scene image into a Scene Interaction Graph (SIG), which specifies the target object, the relevant human body parts, and the intended contact or support edges. Second, an image-editing model inserts one plausible human into the original RGB view while preserving the scene and target object. Third, an agentic contact-region inpainting stage uses the generated human frame and the original scene frame to produce VLM-accepted scene-side contact masks for the body parts named by the SIG. Fourth, these 2D contact masks are projected through the calibrated camera onto visible scene mesh regions, producing one 3D scene contact target per graph edge. In parallel, GVHMR initializes a SMPL-X body from the inpainted human image [7]. The final stage optimizes SMPL-X translation, global orientation, scale, and body pose so that the relevant body contact regions approach the projected scene contact targets while preserving a plausible body and avoiding severe scene penetration.

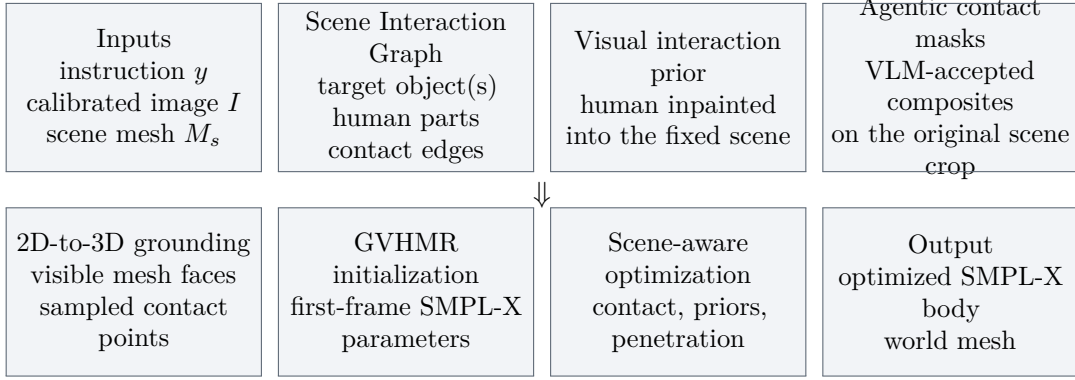


Figure 3.1: Overview of the proposed zero-shot static HSI synthesis pipeline. Foundation models provide semantic and visual interaction priors; camera geometry and scene-aware SMPL-X optimization convert these priors into a metric 3D body placement.

3.2 Problem Formulation

Let $I \in \mathbb{R}^{H \times W \times 3}$ be the selected RGB view of a ScanNet++ scene. The camera is defined by intrinsics K , world-to-camera rotation R , and world-to-camera translation t . The reconstructed scene mesh is denoted

$$M_s = (V_s, F_s), \quad (3.1)$$

where $V_s \in \mathbb{R}^{N_s \times 3}$ are scene vertices in the world coordinate frame and F_s are triangular faces. The instruction y is a free-form text description such as “a person lifting the white trash bin with both hands.” The goal is to estimate SMPL-X parameters

$$\theta = \{\mathbf{T}, \mathbf{r}, \mathbf{p}, \boldsymbol{\beta}, s\}, \quad (3.2)$$

where $\mathbf{T}$ is global translation, $\mathbf{r}$ is global orientation, $\mathbf{p}$ is body pose, $\boldsymbol{\beta}$ is shape, and s is an additional global scale factor. Applying the SMPL-X model to θ produces a human mesh $M_h(\theta) = (V_h(\theta), F_h)$.

The method estimates a body that satisfies three task requirements. First, the body should express the action described by y . Second, the body parts implied by the action should contact the correct target object surface or support surface. Third, the final body should avoid severe scene penetration. Since the task is static, object state changes are outside the scope of the method. For interactions such as opening a drawer or lifting a bin, the synthesized state is interpreted as the contact-ready state before visible object motion begins.

The intermediate Scene Interaction Graph $G = (\mathcal{H}, \mathcal{S}, \mathcal{E})$ makes the interaction constraints explicit. $\mathcal{H}$ contains human body-part nodes, $\mathcal{S}$ contains target scene nodes such as a target object or floor, and $\mathcal{E}$ contains contact/support edges. Each edge $e = (h_e, s_e)$ specifies that

Table 3.1: Example Scene Interaction Graph for the instruction “a person lifting the white trash bin with both hands.”

Field	Example content
Target object	<code>target_object_1</code> : white trash bin
Human part nodes	left hand, right hand, left foot, right foot
Scene nodes	<code>target_object_1</code> , floor
Contact/support edges	left hand $\rightarrow$ target object; right hand $\rightarrow$ target object; left foot $\rightarrow$ floor; right foot $\rightarrow$ floor
Static description	A person is already in contact with the bin and supported by the floor, but the bin remains fixed in the scene.

body part h_e should contact scene region s_e . The synthesis problem can then be viewed as finding θ such that the surface region of $M_h(\theta)$ corresponding to each h_e is geometrically close to the scene surface region corresponding to s_e , and the resulting body does not severely penetrate the scene mesh.

3.3 Scene Interaction Graph and Body Contact Regions

3.3.1 Scene Interaction Graph

The Scene Interaction Graph is the symbolic interface between open-language instructions and metric optimization. Its canonical schema contains four fields:

- target object entries, each with an identifier such as `target_object_1` and a short noun-phrase label;
- human part nodes, written as side-specific body parts such as `person 1, left hand`;
- scene nodes, restricted to active target objects and optionally the floor;
- interaction edges, where each edge maps one human body part to one scene element and includes a short contact note.

The implementation validates the graph before it is used downstream. Target objects must be non-empty, graph edges must reference defined scene nodes, and human parts must come from the supported vocabulary: left hand, right hand, left arm, right arm, left leg, right leg, left foot, right foot, head, hips, and back. This constrained vocabulary is deliberately small. It keeps the graph compatible with the SMPL-X contact-region catalog and avoids introducing language-only body parts that cannot be grounded to mesh vertices.

3.3.2 SMPL-X Contact Regions

SMPL-X provides a dense differentiable body mesh, but its vertices are not directly named by interaction semantics. The method therefore defines a contact-region catalog over the 10,475-vertex SMPL-X body mesh. Each supported graph body part maps to a subset of vertices expected to participate in contact. The catalog includes hand contact regions, foot support regions, and regions for hips, back, and head contacts used in sitting, leaning, lying, and resting interactions.

For a graph edge $e = (h_e, s_e)$, let $B_e \subset \{1, \dots, N_h\}$ be the SMPL-X vertex indices assigned to the human part h_e . During optimization, the current body contact points are

$$P_e^h(\theta) = \{V_h(\theta)_i \mid i \in B_e\}. \quad (3.3)$$

The corresponding scene-side contact region is not manually annotated. It is inferred in image space, projected to the scene mesh, and represented by a sampled 3D point set P_e^s , as described in section 3.5.

3.4 Foundation-Model Visual Reasoning

3.4.1 System Prompts and Symbols

The foundation-model calls are controlled by five system prompts, whose full text is provided in appendix A.1. The SIG system prompt is denoted Π_{SIG} . The human-frame image-editing system prompt is denoted Π_{H} . The VLM system prompt that selects the best human-frame candidate is denoted Π_{sel} ; its final wording is not fixed yet, so the appendix records it as a placeholder. The agentic contact-overlay generation system prompt is denoted Π_{con} . The VLM evaluator system prompt for accepting or rejecting contact masks is denoted Π_{eval} .

3.4.2 SIG Generation

The first model call uses Π_{SIG} to convert the instruction and the selected scene image into a SIG. This system prompt asks the model to identify the target object or objects receiving direct human contact, select only physically important human body parts, include floor support when required, and describe the static contact-ready pose. The graph is then parsed as JSON and validated against the canonical schema. This validation step is important because later stages assume side-specific body parts and valid edge endpoints. If the graph contains unsupported parts, duplicate edges, undefined target objects, or no usable contact edges, the graph is rejected rather than silently propagated.

3.4.3 Human-Frame Synthesis

The second model call uses Π_H and the generated SIG description to edit the original scene image. The image-editing system prompt instructs the model to insert exactly one person performing the interaction while preserving the camera pose, field of view, room geometry, lighting, textures, and all existing objects. The target object must remain fixed: it must not be moved, lifted, opened, rotated, or deformed. This produces human-frame candidates that serve as visual hypotheses for human pose, body orientation, laterality, and approximate contact layout.

These generated frames are not the final output. They are intermediate priors. For human-frame synthesis, sampling multiple edited candidates and selecting the best candidate $\hat{I}$ with the VLM selection system prompt Π_{sel} was empirically sufficient: the selected image mainly needs to provide a plausible action, body orientation, and contact layout for GVHMR initialization. Small visual imperfections in this frame are tolerable because the final metric body is produced only after contact grounding and SMPL-X optimization in the reconstructed scene.

3.4.4 Agentic Contact-Region Inpainting

The contact-region inpainting stage converts the visual interaction hypothesis into scene-side contact masks. It first localizes the generated person in $\hat{I}$ with a SAM3 **person** mask. The resulting human bounding box is padded to include nearby object surfaces and, when needed, expanded to preserve the image aspect ratio. Let this crop window be Ω . The same crop is then applied to two images:

$$I_{\text{ref}} = \text{crop}_{\Omega}(\hat{I}), \quad I_{\text{can}} = \text{crop}_{\Omega}(I). \quad (3.4)$$

Here I_{ref} is the reference crop from the inpainted human image, and I_{can} is the canvas crop from the original scene image. The canvas is therefore not an edited image. It is the original calibrated scene crop containing the empty target object.

The crop also updates the camera intrinsics used downstream. If the crop origin is (x_0, y_0) , the principal point is shifted by this offset while the world-to-camera extrinsics remain unchanged, as formalized in equation (3.9). All contact masks are produced in this cropped coordinate system.

The reference crop I_{ref} is used only to infer which human body parts touch the target object and which object surfaces they touch. The canvas crop I_{can} is the authoritative image for the target object’s pose, geometry, boundaries, appearance, lighting, and pixel coordinate system. Using the canvas crop has two purposes. First, it is easier for the image model to draw contacts on the empty object surface without the human body occluding

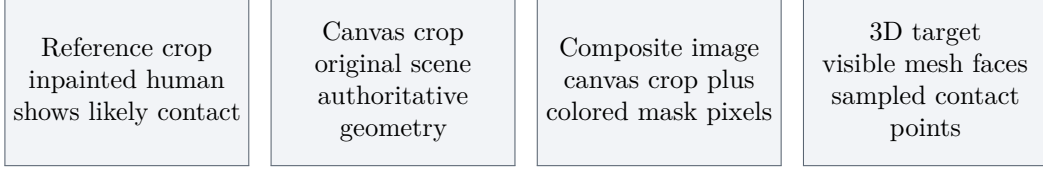


Figure 3.2: Contact-grounding schematic. The generated human is used only to infer where contact should occur. The contact colors are extracted from the generated overlay, pasted onto the original canvas crop to form a composite image, and accepted only after VLM evaluation on that composite.

the object. Second, it prevents image-editing errors in $\hat{I}$ from becoming metric errors. The human-inpainting model may slightly alter the target object’s pose, outline, or appearance. If contact masks were drawn on that altered object and then back-projected, the selected pixels could correspond to the wrong faces on the reconstructed scene mesh. Even visually reasonable contact overlays would then produce wrong 3D contact targets and damage the downstream optimization.

The contact system prompt Π_{con} is instantiated with the SIG and assigns a distinct solid color to each target-object contact body part. Let $\mathcal{P} = \{(h, c_h)\}$ be this palette, where h is a body part and c_h is its RGB color. At agentic round r , the image-editing model receives I_{ref} , I_{can} , $\mathcal{P}$, Π_{con} , and any correction text from the previous round. It outputs a generated contact overlay $O^{(r)}$. The masks in $O^{(r)}$ must be localized contact footprints on the object surface, clipped to the target-object boundary, and must not include the human body silhouette, text labels, arrows, outlines, or changes to any non-mask pixel. This design matters because the optimizer consumes the mask as geometry: a mask on the wrong rung, handle, edge, or side surface creates a wrong 3D target even if the rendered human looks visually plausible.

The raw generated contact overlay $O^{(r)}$ is not trusted as an image of the scene. The image model can still slightly change the target object’s pose, geometry, or appearance while drawing the masks. The implementation therefore extracts only pixels that match colors in $\mathcal{P}$, producing a color-only mask layer

$$Z^{(r)} = \text{extract}_{\mathcal{P}}(O^{(r)}). \quad (3.5)$$

This layer is pasted onto a fresh copy of I_{can} , producing the current composite image

$$C^{(r)} = I_{\text{can}} \oplus Z^{(r)}. \quad (3.6)$$

Here $\oplus$ denotes pixel-wise replacement only at the colored mask pixels. The composite $C^{(r)}$ is the image evaluated by the VLM. It shows where the proposed masks actually land on the true scene crop, rather than where they appear to land on a possibly altered generated overlay.

The composite $C^{(r)}$ is validated before any mask becomes a geometric constraint. A VLM evaluator Φ_{eval} receives three images: I_{ref} , I_{can} , and $C^{(r)}$. It evaluates only the colored masks in the composite:

$$\left(a^{(r)}, \gamma^{(r)}, d^{(r)}\right) = \Phi_{\text{eval}}\left(I_{\text{ref}}, I_{\text{can}}, C^{(r)}; \Pi_{\text{eval}}\right), \quad (3.7)$$

where $a^{(r)}$ is the accept/reject flag, $\gamma^{(r)}$ is the evaluator confidence, and $d^{(r)}$ is the natural-language correction instruction. Evaluating $C^{(r)}$ catches cases where a mask looks correct in $O^{(r)}$ because the target object was subtly changed, but after pixel-wise transfer to the original scene crop the same mask falls on the wrong object part, partly outside the object, or on the background. The evaluator checks for missing contacts, extra contacts, wrong body-part colors, placement on the wrong object surface, shifted masks, and contact footprints that are too large or too small.

If $a^{(r)}$ is false, the correction instruction $d^{(r)}$ is appended to the next contact-inpainting system prompt. The previous composite $C^{(r)}$ is provided only as correction context; the original canvas crop I_{can} remains the required base image for every round. This agentic behavior is important for contact estimation. Independent sampling can repeatedly produce the same type of unusable mask, for example contact regions that are too large or cover the wrong object surface. The evaluator instead tells the image model what is wrong and what should be changed. Because every correction round redraws the masks on a fresh copy of the unedited scene crop and the VLM judges them after transfer back onto that crop, any accidental target-object drift introduced in $O^{(r)}$ cannot accumulate across rounds and cannot propagate into the 2D-to-3D grounding.

Let r^* denote the first accepted round, i.e. $a^{(r^*)} = \text{true}$. The accepted composite $C^{(r^*)}$ is converted into one binary mask per graph edge by nearest-color classification and connected-component filtering:

$$M_e^{2D} = \text{bin}_e\left(C^{(r^*)}, \mathcal{P}\right). \quad (3.8)$$

These accepted masks are the only image-space contact constraints passed to 2D-to-3D grounding.

Floor support is handled separately. When the SIG contains foot-floor edges, SAM3 segments the floor in the original scene view [49]. The floor mask is eroded slightly and used as the support target for the relevant foot contacts. This avoids forcing the image model to localize an arbitrary small foot footprint on a broad support surface when any nearby physically plausible floor support is acceptable.

3.5 2D-to-3D Contact Grounding

The accepted contact masks M_e^{2D} are initially defined in the crop coordinate system, but the optimizer requires 3D scene targets. The method therefore converts each binary mask into visible scene mesh regions using the calibrated ScanNet++ camera.

The crop used for contact-region inpainting determines the projection camera. If the crop begins at pixel (x_0, y_0) , the crop intrinsics are

$$K_{\text{crop}} = \begin{bmatrix} f_x & 0 & c_x - x_0 \\ 0 & f_y & c_y - y_0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (3.9)$$

The camera extrinsics R, t are unchanged by the crop. Each accepted mask M_e^{2D} is interpreted with K_{crop} .

The scene mesh is transformed from world coordinates to the camera coordinate frame using R and t . Only mesh faces visible under the crop camera are considered. For each edge mask M_e^{2D} , candidate scene faces are rasterized or projected into the crop, and faces whose projected pixels overlap the mask are retained. Depth filtering removes faces inconsistent with the visible surface. The implementation uses a depth-jump tolerance, a nearby-depth tolerance, a minimum component size, and a maximum component-gap threshold to avoid selecting disconnected or occluded geometry behind the visible contact region.

The remaining face set is converted into a point cloud. The method samples a fixed number of surface points per edge from the selected scene faces. This point cloud is the scene-side contact target P_e^s . The final edge representation therefore contains the graph nodes, the SMPL-X body contact vertex set B_e , the projected scene face and vertex ids, and the sampled target points P_e^s . Bilateral contacts are spatially disambiguated when necessary so that left and right body parts are paired with the scene-side regions closest to their initialized body locations.

3.6 SMPL-X Initialization and Optimization

3.6.1 GVHMR Initialization

The inpainted human frame $\hat{I}$ provides the only image evidence of the generated person. To initialize a 3D body, the frame is repeated into a short static video clip and passed to GVHMR. GVHMR outputs SMPL-X parameters for the first frame. The first-frame vertices are initially in the camera coordinate frame and are transformed into the ScanNet++ world coordinate frame using the selected image’s COLMAP pose. This initialization gives the

optimizer a plausible full-body pose and approximate camera-space placement, but it is not assumed to satisfy the interaction constraints exactly.

3.6.2 Optimized Variables

The scene mesh, camera calibration, and graph-derived contact targets are fixed during optimization. The optimized body variables are translation, global orientation, body pose, and global scale. Shape parameters are initialized from GVHMR and kept as part of the recovered body state, while the additional scale variable allows the canonical SMPL-X height to fit a reasonable human-height range. The optimizer exports the world-frame mesh together with the optimized parameter file.

3.6.3 Objective

The optimization objective combines graph-derived contact terms, a penetration penalty, and priors that keep the body close to the GVHMR initialization:

$$\begin{aligned} \mathcal{L}(\theta) = & \lambda_{\text{contact}} \mathcal{L}_{\text{contact}} + \lambda_{\text{penetration}} \mathcal{L}_{\text{penetration}} + \lambda_{\text{pose}} \mathcal{L}_{\text{pose}} + \lambda_{\text{orient}} \mathcal{L}_{\text{orient}} \\ & + \lambda_{\text{height}} \mathcal{L}_{\text{height}}. \end{aligned} \quad (3.10)$$

The contact loss attracts the selected SMPL-X contact vertices to the projected scene contact points:

$$\mathcal{L}_{\text{contact}} = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} d(P_e^h(\theta), P_e^s), \quad (3.11)$$

where $d(\cdot, \cdot)$ is a nearest-neighbor point-cloud distance with an edge-specific reduction. The implementation reports the square root of this value as a contact distance in meters for interpretability.

The penetration loss samples points on the scene surface and queries their signed distance to the posed SMPL-X body using VolumetricSMPL [55]. Let q_j be sampled scene points and $\text{SDF}_{h,\theta}(q_j)$ be their signed distance to the human body. With clearance margin m ,

$$\mathcal{L}_{\text{penetration}} = \frac{1}{|\mathcal{Q}^+|} \sum_{q_j \in \mathcal{Q}^+} \max(0, m - \text{SDF}_{h,\theta}(q_j)), \quad \mathcal{Q}^+ = \{q_j \mid m - \text{SDF}_{h,\theta}(q_j) > 0\}. \quad (3.12)$$

This term penalizes scene points that lie inside the body or within the configured clearance margin.

The pose prior penalizes deviations from the GVHMR body pose. The orientation prior penalizes the relative axis-angle rotation between the current global orientation and the initialization. The height prior constrains the scaled canonical body height to a plausible interval around a target height.

Table 3.2: Optimization terms used by the scene-aware SMPL-X fitting stage.

Term	Status	Purpose
$\mathcal{L}_{\text{contact}}$	default	Pulls each body contact region toward the corresponding projected scene contact target.
$\mathcal{L}_{\text{penetration}}$	default	Penalizes scene points inside or too close to the posed SMPL-X body.
$\mathcal{L}_{\text{pose}}$	default	Keeps the optimized body pose close to the GVHMR initialization.
$\mathcal{L}_{\text{orient}}$	default	Prevents implausible global rotations away from the initialized body orientation.
$\mathcal{L}_{\text{height}}$	default	Keeps the scaled body within a plausible human-height range.

3.6.4 Optimization Schedule

The optimizer uses Adam with a typical learning rate of 10^{-3} . A typical run uses 2000 iterations. The optimization is stage-wise. In the rigid stage, translation and scale are trainable while global orientation and body pose are frozen. Empirically, the GVHMR initialization provides a reliable global body orientation for the calibrated input views, so the first stage preserves this orientation while aligning the body to the scene contact targets. In the pose stage, translation, global orientation, scale, and body pose are all trainable. This allows the body to refine its local articulation while remaining regularized by the GVHMR pose and orientation priors.

The implementation samples approximately 700,000 scene surface points for the penetration loss in a typical run and samples 2048 scene contact points per graph edge. It saves per-iteration loss logs, depth visualizations, penetration debug artifacts, optimized SMPL-X parameters, and the final world-coordinate mesh.

3.7 Implementation Details and Artifacts

The implementation combines ScanNet++ camera and mesh data, SAM3 concept-prompted segmentation, Gemini-style image editing for human insertion and agentic contact-region inpainting, a Gemini VLM evaluator for contact-mask acceptance, GVHMR for monocular SMPL-X initialization, SMPL-X for the differentiable body model, VolumetricSMPL for body signed-distance queries, and PyTorch/PyTorch3D utilities for optimization, projection, and rasterization. Blender is used later to render optimized interactions for qualitative inspection and semantic evaluation, but rendering and evaluation metrics are part of the experimental protocol in chapter 4, not the core synthesis method.

This chapter intentionally stops at the production of the optimized static 3D human-

Table 3.3: Main intermediate and final artifacts produced by the methodology pipeline.

Artifact	Role in the pipeline
<code>sig.json</code>	Validated Scene Interaction Graph with target object labels, body parts, and contact/support edges.
<code>inpainted_frame_resized.png</code>	Image-editing output resized to the original calibrated view; used for contact inference and GVHMR initialization.
<code>rounds/round_*/</code>	Agentic contact-inpainting system-prompt packages, generated overlays $O^{(r)}$, color layers $Z^{(r)}$, composites $C^{(r)}$, and VLM evaluations $(a^{(r)}, \gamma^{(r)}, d^{(r)})$.
<code>agentic_summary.json</code>	Contact-mask acceptance state, accepted round r^* , palette $\mathcal{P}$, target object label, and latest VLM evaluation.
<code>contact_overlay.png</code>	Accepted composite $C^{(r^*)}$: original canvas crop with contact-color pixels pasted onto it.
<code>contact_masks/*.png</code>	Binary masks M_e^{2D} corresponding to the accepted composite for each body-part contact edge in the cropped canvas coordinate system.
<code>contact_spec.json</code>	Crop-adjusted camera intrinsics K_{crop} and color palette $\mathcal{P}$ for contact-mask extraction.
<code>hmr4d_results.pt</code>	GVHMR output containing the initial SMPL-X parameters.
<code>optimized_frame_0000.pt</code>	Optimized SMPL-X translation, orientation, body pose, shape, and scale.
<code>frame_0000_world.ply</code>	Final human mesh transformed to the ScanNet++ world coordinate frame.

scene interaction. Dataset selection, interaction counts, baselines, quantitative metrics, rendered-view evaluation, and numerical results are defined in the following chapters.

Chapter 4

Experiments

This chapter defines the experimental protocol used to validate the thesis claims.

4.1 Research Questions

State the questions the experiments answer. Each question should map to a table, figure, ablation, or qualitative analysis.

4.2 Datasets

Describe all datasets, splits, annotations, preprocessing steps, and any generated data.

4.3 Baselines

List baseline methods and explain how they were configured. Make comparisons fair and reproducible.

4.4 Metrics

Define quantitative metrics and explain what each metric captures. Include limitations of each metric where relevant.

Table 4.1: Example experiment configuration table. Replace with the final setup.

Item	Value
Dataset split	train / validation / test
Batch size	–
Optimizer	–
Hardware	–

4.5 Experimental Setup

Specify training schedule, optimizer, learning rate, batch size, number of runs, random seeds, hardware, and software versions.

Chapter 5

Results and Discussion

This chapter reports the main results, ablations, qualitative examples, and limitations.

5.1 Main Results

Present the primary quantitative results first. Explain the trend, not just the numbers.

5.2 Ablation Studies

Isolate the effect of each contribution. Keep ablations tied to the claims made in the introduction.

5.3 Qualitative Results

Show representative successes and failures. Qualitative figures should make the method’s behavior inspectable.

5.4 Limitations

Discuss failure cases, data bias, compute cost, assumptions, and evaluation gaps. Strong limitations make the thesis more credible.

Table 5.1: Main result table placeholder.

Method	Metric 1	Metric 2	Metric 3
Baseline	–	–	–
Proposed	–	–	–

5.5 Discussion

Connect the results back to the motivation. Explain what the experiments imply for the broader research problem.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

Summarize the thesis in a few paragraphs: problem, approach, evidence, and main takeaway. Avoid introducing new experiments here.

6.2 Future Work

Describe concrete extensions that follow from the limitations and results. Separate near-term engineering improvements from longer-term research directions.

6.3 Closing Remarks

End with a brief statement about the expected impact or usefulness of the work.

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Appendix A

Additional Material

A.1 System Prompts

This appendix records the system prompts used by the foundation-model stages in chapter 3. The symbols below are used in the methodology chapter.

Table A.1: System-prompt symbols used in the methodology.

Symbol	Stage	Source or status
Π_{SIG}	SIG generation	4DHSI/01_Generate_SIG/system_prompt_sig.md
Π_{H}	Human-frame inpainting	4DHSI/02_Generate_Human_Frame/system_prompt_human_inpaint.md
Π_{sel}	Best human-frame selection	Placeholder; final system prompt not fixed yet.
Π_{con}	Contact overlay generation	4DHSI/03_Estimate_Contact_Agentic/system_prompt_estimate_contact_agentic.md
Π_{eval}	Contact VLM evaluation	4DHSI/03_Estimate_Contact_Agentic/vlm_prompt_evaluate_contact.md

A.1.1 SIG Generation System Prompt Π_{SIG}

You generate a Scene Interaction Graph (SIG) for a static human-scene interaction.

Produce a compact graph containing the target object or objects, relevant human body parts, direct physical contact edges, and a static interaction description.

Input:

- A scene image from the selected camera view.
- A JSON request:

```
{
  "interaction": "short interaction description"
}
```

Output only valid JSON. Do not include markdown or commentary.

Required output schema:

```
{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "short primary target object name"
    },
    {
      "id": "target_object_2",
      "label": "short secondary target object name"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand"
  ],
  "scene_nodes": [
    "target_object_1",
    "target_object_2",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "brief reason for this contact"
    }
  ],
  "interaction": "Static interaction description under 150 words."
}
```

Rules:

1. Use the interaction text to identify the intended human-scene contact, and use the
↪ image only to understand visible object context.
2. `target\_objects` should contain the scene object that receive direct human contact
↪ in the static pose.
3. Use one target object when the contact is with a single object or multiple
↪ structural parts of the same object, such as a chair back, bed frame, or bathtub
↪ side.
4. Use two target objects only when two separate scene objects are directly contacted.
↪ For example, use "bathtub" and "curtain rail" when one foot contacts the bathtub
↪ and one hand contacts the curtain rail.
5. Keep target object labels short noun phrases.
6. Use human part nodes only from this vocabulary:
 - left hand
 - right hand
 - left arm
 - right arm
 - left leg
 - right leg
 - left foot
 - right foot
 - head
 - hips
 - back

7. Include a human part node only when that part is physically important for contact
 ↳ or pose.
8. `scene\_nodes` can include only:
 - target_object_1
 - target_object_2
 - floor
9. Include `target\_object\_2` only if it exists in `target\_objects` and has at least
 ↳ one direct contact edge.
10. `interaction\_edges` represent direct physical contact only.
11. Use `scene\_element: "target\_object\_1"` or `scene\_element: "target\_object\_2"` for
 ↳ contacts with the corresponding target object.
12. Use floor edges only for required direct floor support. If there is no floor
 ↳ contact, simply omit `floor` from `scene\_nodes` and `interaction\_edges`.
13. Do not create object-part nodes.
14. Keep interaction edges focused on physically necessary contacts.
15. The output `interaction` must describe the static pose and physical contacts under
 ↳ 150 words. It must explicitly restate every `interaction\_edges` contact using the
 ↳ same side-specific human part names, such as "left hand", "right foot", or "back",
 ↳ and the contacted scene object label or floor. Avoid vague phrases such as "one
 ↳ hand", "both feet", or "legs" when the edge uses specific left/right parts. It
 ↳ should also describe clothing, shoes, hair, and facial expression.

Few-shot examples:

These few-shot examples are written without a scene image. They demonstrate the expected JSON structure and explicit contact wording. For the real request, use the provided scene image to decide the exact contact configuration, including which side-specific hand, foot, leg, or arm is in contact.

Example 1 input JSON:

```
{
  "interaction": "a person lifting the brown small wooden table with both hands"
}
```

Example 1 output:

```
{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "brown small wooden table"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand",
    "person 1, left foot",
    "person 1, right foot"
  ],
  "scene_nodes": [
    "target_object_1",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "the left hand grips the table while preparing to lift"
    },
    {
      "human_part": "right hand",
      "scene_element": "target_object_1",

```

```

    "notes": "the right hand grips the opposite side of the table"
  },
  {
    "human_part": "left foot",
    "scene_element": "floor",
    "notes": "the left foot supports the standing pose on the floor"
  },
  {
    "human_part": "right foot",
    "scene_element": "floor",
    "notes": "the right foot supports the standing pose on the floor"
  }
],
"interaction": "A person stands beside the brown small wooden table in a
↪ forward-leaning pose. The left hand grips one side of the brown small wooden
↪ table, the right hand grips the opposite side, the left foot is planted on the
↪ floor, and the right foot is planted on the floor. The person has short black
↪ hair, a neutral focused facial expression, a gray shirt, blue jeans, and white
↪ sneakers."
}

```

Example 2 input JSON:

```

{
  "interaction": "a person lying on bed with his head on the pillow"
}

```

Example 2 output:

```

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "bed with pillows"
    }
  ],
  "human_part_nodes": [
    "person 1, left leg",
    "person 1, right leg",
    "person 1, head",
    "person 1, hips",
    "person 1, back"
  ],
  "scene_nodes": [
    "target_object_1"
  ],
  "interaction_edges": [
    {
      "human_part": "head",
      "scene_element": "target_object_1",
      "notes": "the head rests on the pillow as part of the bed setup"
    },
    {
      "human_part": "hips",
      "scene_element": "target_object_1",
      "notes": "the hips are supported by the mattress"
    },
    {
      "human_part": "back",
      "scene_element": "target_object_1",
      "notes": "the back rests against the mattress"
    }
  ],
  {

```

```

    "human_part": "left leg",
    "scene_element": "target_object_1",
    "notes": "the left leg rests on the bed"
  },
  {
    "human_part": "right leg",
    "scene_element": "target_object_1",
    "notes": "the right leg rests on the bed"
  }
],
"interaction": "A person lies on the bed with pillows in a relaxed static pose. The
↪ head rests on a pillow, the back rests on the mattress, the hips are supported
↪ by the mattress, the left leg rests on the bed, and the right leg rests on the
↪ bed. The person has short black hair, a calm sleepy facial expression, a gray
↪ shirt, soft pants, and socks."
}

```

Example 3 input JSON:

```

{
  "interaction": "a person stepping over the bathtub wall, one foot inside the tub,
  ↪ the other foot still on the floor, one hand holding the curtain rail for balance"
}

```

Example 3 output:

```

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "bathtub"
    },
    {
      "id": "target_object_2",
      "label": "curtain rail"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, left foot",
    "person 1, right foot"
  ],
  "scene_nodes": [
    "target_object_1",
    "target_object_2",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_2",
      "notes": "the left hand grips the curtain rail for balance"
    },
    {
      "human_part": "left foot",
      "scene_element": "target_object_1",
      "notes": "the left foot presses inside the bathtub while stepping over the wall"
    },
    {
      "human_part": "right foot",
      "scene_element": "floor",
      "notes": "the right foot stays planted on the floor outside the bathtub"
    }
  ]
}

```

```

],
"interaction": "A person steps over the bathtub wall in a careful balancing pose.
↳ The left hand grips the curtain rail, the left foot presses inside the bathtub,
↳ and the right foot is planted on the floor outside the tub. The person has short
↳ black hair, a focused facial expression, a gray shirt, rolled-up pants, and bare
↳ feet."
}

```

Example 4 input JSON:

```

{
  "interaction": "a person bending down in front of the lower drawer, one hand pulling
↳ the cabinet door open and the other hand holding the counter edge for balance"
}

```

Example 4 output:

```

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "lower cabinet door"
    },
    {
      "id": "target_object_2",
      "label": "counter edge"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand",
    "person 1, left foot",
    "person 1, right foot",
    "person 1, hips"
  ],
  "scene_nodes": [
    "target_object_1",
    "target_object_2",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "the left hand pulls the lower cabinet door"
    },
    {
      "human_part": "right hand",
      "scene_element": "target_object_2",
      "notes": "the right hand holds the counter edge for balance"
    },
    {
      "human_part": "left foot",
      "scene_element": "floor",
      "notes": "the left foot supports the squat on the floor"
    },
    {
      "human_part": "right foot",
      "scene_element": "floor",
      "notes": "the right foot supports the squat on the floor"
    }
  ],
}

```



```

"interaction": "A person bends down in front of the lower cabinet door in a balanced
↪ pose. The left hand pulls the lower cabinet door, the right hand holds the
↪ counter edge, the left foot is planted on the floor, and the right foot is
↪ planted on the floor. The person has short black hair, a neutral concentrated
↪ facial expression, a gray shirt, blue jeans, and white sneakers."
}

```

Example 5 input JSON:

```

{
  "interaction": "a person hanging from the pull-up bar with both hands, elbows bent,
  ↪ knees tucked upward, and both feet off the floor"
}

```

Example 5 output:

```

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "pull-up bar"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand"
  ],
  "scene_nodes": [
    "target_object_1"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "the left hand grips the pull-up bar"
    },
    {
      "human_part": "right hand",
      "scene_element": "target_object_1",
      "notes": "the right hand grips the pull-up bar"
    }
  ],
  "interaction": "A person hangs from the pull-up bar with elbows bent and knees
  ↪ tucked upward. The left hand grips the pull-up bar, the right hand grips the
  ↪ pull-up bar, the left foot stays off the floor, and the right foot stays off the
  ↪ floor. The person has short black hair, a strained focused facial expression, a
  ↪ gray athletic shirt, dark shorts, and sneakers."
}

```

A.1.2 Human-Frame Inpainting System Prompt Π_H

Input Image:

- The image is the original scene. Use the interaction description to identify the
 ↪ existing target object that must remain fixed and contacted.

Scene Preservation Instructions:

- Use the provided scene image as a fixed camera reference.

- Insert exactly one person performing the described interaction with the target
↪ object.
- This frame must represent the pre-motion start state: the person is already in
↪ natural contact

with the target object of the scene, but no visible object motion has begun yet.

- Target object is already present in the scene. Do not add it.
- Preserve the camera pose, field of view, perspective, room geometry, lighting,
↪ shadows, textures,
and all existing object identities.
- Do not add new furniture or remove existing items.
- Keep the target object in exactly the same pose, position, orientation, scale,
↪ material,
and appearance as in the original scene.
- Do not move, rotate, deform, lift, translate, or reposition the target object in any
↪ way.
- Only add the human in plausible contact, as if about to start the action in the next
↪ frame.

Output:

- Return only the edited scene image.

A.1.3 Best Human-Frame Selection System Prompt Π_{sel}

PLACEHOLDER: Final best-frame selection system prompt not fixed yet.

Intended role:

- Given the original scene image, the interaction description or SIG description, and
↪ multiple human-inpainted candidate frames, select the single candidate used as the
↪ visual prior for GVHMR initialization and contact reasoning.

Intended ranking criteria:

1. Action expression: the human should clearly express the requested interaction or
↪ contact-ready pre-action state.
2. Target-object preservation: the target object should remain fixed in pose,
↪ location, scale, material, and appearance relative to the original scene.
3. Plausible body pose: the inserted human should have a physically plausible
↪ full-body pose, orientation, scale, and support.
4. Contact readiness: the relevant body parts should appear close to or in natural
↪ contact with the intended target object surfaces or floor support regions.

Intended output:

- Return the index or identifier of the best candidate and a concise reason for the
↪ selection.

A.1.4 Agentic Contact-Overlay System Prompt Π_{con}

Provided Images:

Reference Image: An image showing a human interacting with the target object. Use this
↪ image only to infer which human body parts are physically contacting the target
↪ object and which object-surface regions are being touched.

Canvas Image: An image of the same scene without the human. Use this image as the base
↪ image and as the authoritative reference for the target object's final position,
↪ pose, geometry, and boundaries.

Task Description:

Generate an edited copy of the Canvas Image with solid-color segmentation masks
↪ overlaid only on the target object surface regions that are contacted by the human
↪ in the Reference Image.

Infer the contacted region relative to the target object in the Reference Image, then
↪ place the corresponding contact mask on the matching object region in the Canvas
↪ Image.

Critical Base-Image Rule:

Use the Canvas Image as the only valid base image. Preserve the Canvas Image exactly
↪ and only add the specified solid-color contact masks. Do not use the Reference
↪ Image as the base. Do not redraw, regenerate, stylize, relight, crop, pad, resize,
↪ move, or otherwise alter the Canvas Image.

Analysis Requirements:

1. Contact Inference:

Analyze the Reference Image to determine which specified human body parts are
↪ physically touching the target object.

2. Object-Region Identification:

For each contacting body part, identify the specific region of the target object
↪ being touched, such as the tabletop front edge, tabletop side edge, tabletop
↪ upper surface, tabletop underside, table leg, chair seat, chair arm, cabinet
↪ handle, ladder rung, rail, shelf, handle, edge, or other relevant object part.

3. Object-Relative Localization:

Map each inferred contact region onto the corresponding part of the target object
↪ in the Canvas Image. Use the Canvas Image object pose and boundaries as ground
↪ truth. Prioritize correct placement on the object in the Canvas Image over
↪ exact pixel alignment with the Reference Image.

4. Contact Footprint Generation:

Create a localized contact mask only on the object surface where contact occurs.
↪ The mask should represent the apparent contact footprint on the object, not the
↪ full visible human body part.

Execution Requirements:

Base Image:

Return the Canvas Image with the contact masks overlaid on top of it.

Canvas Size:

Match the Canvas Image dimensions and coordinate system.

Mask Placement:

Place each mask on the corresponding target-object region in the Canvas Image, using
↪ object-relative correspondence rather than direct coordinate transfer.

Mask Shape:

Use precise, dense, solid-color segmentation masks shaped like the localized contact
↪ footprint on the object surface. The mask should reflect the approximate size,
↪ orientation, and extent of the contact implied by the touching body part.

Object-Surface Clipping:

All mask pixels must be clipped strictly to the target object boundary in the Canvas
↪ Image. No mask may extend onto the human body, floor, wall, background, or any
↪ non-target-object region.

Scene Preservation:

Use the Canvas Image as a fixed camera reference. Preserve the camera pose, field of
↪ view, perspective, resolution, aspect ratio, room geometry, lighting, shadows,
↪ textures, and all existing object identities.

Keep the target object in exactly the same pose, position, orientation, scale,
↪ material, and appearance as in the Canvas Image. Do not move, rotate, deform,
↪ lift, translate, resize, crop, pad, or reposition the target object or any other
↪ scene element in any way.

Only add the specified solid-color contact mask overlays. Leave every non-overlay
↪ pixel visually unchanged from the Canvas Image.

Colors:

Use only the exact specified solid colors for the contacting body-part masks. Do not
↪ use transparency, antialiasing gradients, shadows, blended colors, or pastel
↪ variants.

Restrictions:

Do not add the generated human body.

Do not copy or paste the full visible hand, foot, hip, arm, leg, or other body-part
↪ silhouette.

Do not transfer masks by raw Reference Image pixel coordinates.

Do not add text labels, arrows, legends, boxes, keypoints, outlines, or any other
↪ annotations.

Show only the original Canvas Image with solid localized contact masks overlaid on the
↪ target object.

Example Guidance:

If the Reference Image shows a person gripping a ladder, infer which rungs or rails
↪ the hands and feet contact. Then place solid masks on those same ladder regions in
↪ the Canvas Image, aligned to the ladder as it appears in the Canvas Image.

A.1.5 Contact VLM Evaluation System Prompt Π_{eval}

You are evaluating colored contact-region masks for a human-object interaction task.

You are given three images:

Image 1: Reference Image.

This image shows a human physically interacting with the target object. Use it to
↪ infer which body parts are touching which object surfaces.

Image 2: Original Canvas Image.

This image shows the same scene without the human. This is the authoritative geometry
↪ and base image.

Image 3: Current Composite Image.

This is the original Canvas image with colored contact masks pasted onto it. Evaluate
↪ only the colored masks.

Target object:

{target_object}

Color mapping:

{color_mapping}

Task:

Decide whether the colored masks in the Current Composite Image correctly mark the
↪ localized contact regions implied by the Reference Image.

Focus especially on location. For each mask, check whether it is on the correct object
↪ part and correct local surface.

Evaluate these possible errors:

1. Missing contact: a body part touches the target object in the Reference Image but
↪ its mask is absent.
2. Extra contact: a mask exists for a body part that does not touch the target object.
3. Wrong color: a contact uses the wrong color for the body part.
4. Wrong object surface: the mask is on the wrong object part, such as wrong rung,
↪ wrong rail, wrong edge, wrong handle, wrong shelf, wrong tabletop region, etc.
5. Shifted mask: the mask is too far left, right, up, or down from the correct contact
↪ location.
6. Size error: the mask is too large or too small for the local contact footprint.

For every problem, write a concrete correction instruction for the next ChatGPT
↪ image-generation prompt.

Good correction examples:

- "Keep the original canvas unchanged and move only the green right-foot mask upward
↪ onto the lower ladder rung."
- "The yellow left-hand mask is missing. Add a small yellow mask on the vertical side
↪ rail where the left hand contacts the ladder."
- "The blue foot mask is on the wrong rung. Move it down to the rung directly under
↪ the visible foot contact."
- "The red right-hand mask is too large. Keep its center but make it smaller and more
↪ focused."
- "This mask is correct. Keep it unchanged."

Bad correction examples:

- "The result is bad."
- "Fix the mask."
- "Try again."

Return strictly valid JSON only:

```
```json
{
 "done": false,
 "confidence": 0.0,
 "correction_instruction": "text"
}
```
```

Set "done": true only if all required contacts are present, there are no extra masks,
↪ each mask uses the correct color, each mask is on the correct target-object
↪ surface, and each mask has acceptable location and size.